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Topological Quantum Computing

Exploring Majorana Zero Modes or something subtitley

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Abstract

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Contents

1 Introduction

Quantum computing offers the possibility of solving classes of problems that are intractable on classical hardware, but this promise is limited by one central difficulty: quantum states are fragile. Decoherence, noise, and imperfect control make it challenging to preserve and manipulate quantum information long enough to perform useful computations. One of the most attractive proposed solutions to this problem is topological quantum computing, where information is stored non-locally in collective quantum states that are intrinsically less sensitive to local perturbations. In this picture, protection is not achieved only through better engineering, but through the structure of the quantum states themselves.

Majorana zero modes have therefore attracted broad interest as candidate building blocks for fault-tolerant quantum computing. In condensed-matter systems, they appear not as fundamental particles, but as emergent zero-energy quasiparticles in topological superconductors [2]. Their most important feature is that a pair of spatially separated Majorana modes can encode a non-local fermionic degree of freedom. Because local perturbations cannot easily access this information, such states offer a route toward more robust quantum operations. In addition, Majorana modes are predicted to obey non-Abelian exchange statistics, meaning that exchanging them can implement quantum gates through braiding rather than through purely local control pulses.

Although the basic theoretical picture is well established, realizing and controlling Majorana modes in realistic devices remains a major challenge. Much of the experimental and theoretical effort has therefore focused on engineered hybrid systems, where conventional superconductivity, tunneling, spin structure, and confinement can be combined to produce effective topological behavior [1]. Among the simplest of these platforms is the quantum-dot–superconductor–quantum-dot setup, often referred to as the Poor Man's Majorana system. This device can be understood as a minimal analogue of the Kitaev chain: hopping between dots plays the role of the kinetic term, non-local pairing induced by the superconductor plays the role of effective p-wave pairing, and gate-controlled dot energies act as local chemical potentials.

The appeal of this platform is twofold. First, it is conceptually simple enough to connect directly to the Kitaev-chain picture, which makes it well suited for theoretical analysis. Second, it is experimentally tunable: dot energies, effective pairing strengths, and tunnel couplings can all be adjusted in a controlled way. At the same time, this simplicity comes with limitations. In finite and interacting systems, one should be careful not to overstate the meaning of low-energy degeneracies or Majorana-like observables. The relevant question is therefore not only whether a model displays signatures associated with Majorana physics, but whether those signatures survive under interactions and whether they remain useful for braiding operations.

This thesis studies that question in interacting chains of coupled quantum dots. More specifically, I investigate whether an interacting n-dot Poor Man's Majorana chain can be numerically optimized to produce low-energy states with the expected Majorana-like characteristics, and whether those optimized states support a useful braiding protocol. The goal is not to claim exact topological protection in a finite microscopic model. Rather, the aim is to identify parameter regimes where the low-energy structure is sufficiently close to the Majorana ideal to make the physical interpretation meaningful and the braiding behavior nontrivial.

To do this, I model the system with an interacting quantum-dot Hamiltonian containing on-site energies, nearest-neighbor hopping, induced pairing, and Coulomb repulsion. The search for promising parameter regimes is formulated as an optimization problem, where the loss function is built from several physically motivated Majorana criteria: near-degeneracy between parity sectors, finite low-energy gaps, small charge differences between parity partners, and a Majorana-polarization profile consistent with edge-localized modes. Once optimized configurations are found, I analyze their parity-resolved spectra and local diagnostics, reconstruct effective Majorana operators from the low-energy manifold, and compare an ideal four-Majorana braiding protocol with a braid generated by projected microscopic junction operators.

The main results show a clear distinction between weakly and strongly interacting regimes. The optimization reproduces the known non-interacting two-dot sweet spot and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. Among the interacting cases studied here, the three-dot system at weak interaction emerges as the most convincing candidate: it combines a very small ground-state splitting, a clear gap to excited states, and a spatial Majorana profile concentrated at the chain edges. In the lowest projected manifold, the corresponding microscopic braid agrees remarkably well with the ideal effective-model braid. However, this agreement deteriorates as stronger interactions and higher-energy sectors are included, indicating that useful braiding depends sensitively on how well the exchange remains confined to a clean low-energy subspace.

The broader motivation of the thesis is therefore not only to identify candidate Majorana-like configurations, but also to clarify the boundary between effective low-energy success and microscopic breakdown. In that sense, the work is positioned between abstract Majorana models and realistic device-level questions: the optimization tells us which interacting parameter regimes are promising, while the braiding analysis tests whether those regimes support the operation that ultimately matters for topological quantum computing.

The remainder of the thesis is organized as follows. Chapter 2 presents the theoretical background, beginning with superconductivity and the Bogoliubov-de Gennes formalism, then introducing the Kitaev chain, the quantum-dot realization of Poor Man's Majoranas, and the basic logic of braiding. Chapter 3 describes the numerical methods used to optimize the interacting dot chains, analyze the resulting low-energy states, and construct the braiding protocol. Chapter 4 presents the optimization results, local Majorana diagnostics, and braiding benchmarks in both the idealized and projected microscopic models. Chapter 5 discusses the physical interpretation of these findings, their limitations, and the role of interactions in shaping the low-energy manifold. Finally, Chapter 6 summarizes the main conclusions and outlines possible directions for future work.

2 Theory

2.1 Majorana Fermion

Quick introduction to Majorana fermions, their properties. Discuss the concept of Majorana zero modes (MZMs) and their significance in topological quantum computing.

2.2 Topological Quantum Computing

Introduce the concept of topological quantum computing, emphasizing the role of MZMs in encoding and manipulating quantum information. Discuss the advantages of topological quantum computing, such as robustness against local perturbations and decoherence.

2.3 The Superconducting State

2.3.1 From Cooper Pairing to the Energy Gap

Superconductivity arises when electrons in a metal form bound pairs, known as Cooper pairs, below a critical temperature T_c. These pairs condense into a coherent quantum state that can carry current without resistance, and also expel magnetic fields via the Meissner effect. This immediately distinguishes a superconductor from a perfect conductor, because it is a distinct thermodynamic phase with long-range quantum coherence [4].

The microscopic origin of superconductivity lies in an effective attractive interaction between electrons, mediated by phonons (lattice vibrations). It is somewhat counterintuitive that electrons, which naturally repel each other due to their negative charge, can experience an effective attraction. However, when an electron moves through a lattice structure of positive ions, it distorts the lattice, pulling a group of the ions towards itself. This distortion creates a region of increased positive charge density that can attract another electron with opposite momentum and spin, leading to the formation of a Cooper pair [5].

Even a weak attraction can destabilize the normal metallic Fermi sea, allowing electrons of opposite momenta and spins to form bound pairs. These pairs behave as bosons (integer spin particles) and condense into a macroscopic quantum state described by a collective wavefunction $\psi(r)$, whose phase coherence gives rise to the supercurrent.

At the mean-field level, this collective behaviour opens an energy gap Δ around the Fermi surface. This gap represents the minimum energy needed to break a Cooper pair and create excitations, called Bogoliubov quasiparticles, that are mixtures of electron and hole states. The presence of this gap explains the vanishing resistivity and the exponential suppression of heat capacity and other thermodynamic quantities at low temperatures [4], [3].

The pairing gap is what makes superconductors so special for topological applications. Because single-particle excitations cost a finite energy, the superconducting condensate is robust against small perturbations and other local noise. When combined with spin-orbit coupling and appropriate confinement or hybridization, this same pairing mechanism can give rise to topological superconductivity, where the quasiparticles at zero energy behave as Majorana modes.

In essence, superconductivity provides both the microscopic pairing mechanism and the macroscopic phase coherence needed to host non-local, topologically protected excitations, making it the natural foundation for exploring Majorana physics in condensed matter systems.

2.3.2 Bogoliubov-de Gennes Formalism

While the BCS theory captures the essence of superconductivity as a condensate of Cooper pairs, it assumes a spatially uniform order parameter Δ. In real systems, superconductivity can vary across space due to interfaces, impurities, or confinement. To describe such spatially inhomogeneous superconductors, the BCS framework must be generalized. This leads to the

Bogoliubov-de Gennes (BdG) formalism [4], [3].

$$\begin{pmatrix} H_0(r) & \Delta(r) \\ \Delta^*(r) & -H_0^*(r) \end{pmatrix} \begin{pmatrix} u_n(r) \\ v_n(r) \end{pmatrix} = E_n \begin{pmatrix} u_n(r) \\ v_n(r) \end{pmatrix} \tag{1}$$

where $H_0 = \frac{p^2}{2m} - \mu + V(r)$ describes the normal-state single particle Hamiltonian, and $\Delta(r)$ is the superconducting pair potential.
Each solution of the BdG equations (1) corresponds to a Bogoliubov quasiparticle, a coherent superposition of electron and hole amplitudes $(u_n(r), v_n(r))$. The resulting excitation spectrum is symmetric about zero energy due to the intrinsic particle-hole symmetry of the BdG Hamiltonian:

$$\Xi H_{BdG} \Xi^{-1} = -H_{BdG}, \quad \Xi = \tau_x K \tag{2}$$

where τ_x acts in particle-hole space and K is complex conjugation. This ensures that if E_n is an eigenvalue, so is $-E_n$.
This symmetry has an important consequence: superconductors no longer conserve particle number, only fermion parity, and the eigenstate at $\pm E_n$ correspond to the same physical excitation. A zero-energy solution ($E_n = 0$) indicates a self-conjugate quasiparticle, $\gamma = \gamma^\dagger$, which is the defining property of a Majorana mode [2].
The BdG formalism connects the microscopic BCS theory and the modern description of topological superconductivity. It provides a natural framework for modeling hybrid nanostructures, such as quantum dots coupled to superconducting leads, and low-dimensional systems where topological phases can emerge.

2.3.3 Hybrid Semiconductor-Supercondcutor Systems

Superconductivity on its own already provides a platform for charge transport without resistance, and for pairing electrons into coherent states. However, when a conventional superconductor is coupled to another material, such as a semiconductor, it can transfer its essential properties through what is known as the *proximity effect.* This hybridization allows us to engineer superconductivity in systems that do not exhibit it intrinsically, and forms an experimental foundation for realizing Majorana modes in condensed matter systems [1].

(i) Hybdird Systems: The Proximity Effect and Induced Superconductivity At the interface between a semiconductor and a superconductor, the superconducting condensate can penetrate into the semiconductor, through a process called the proximity effect. As a result, electrons in the semiconductor acquire pairing correlations from the superconductor, giving rise to an induced superconducting gap Δ_{ind}, even though the semiconductor itself lacks an intrinsic pairing interaction.
This induced pairing allows the semiconductor to behave as a proximitized superconductor. However, the advantage of using a semiconductor is not just that it becomes superconducting, but that it is highly tunable and has strong spin-related effects that are difficult to achieve in conventional superconductors.
In order to induce a topological transition, the idea is to exploit the competition between three different effects. First, the s-wave superconducting proximity effect provides the pairing mechanism. Second, the Zeeman energy $E_Z = \frac{1}{2} g \mu_b B$, can break Cooper pairs by aligning their electron spins. Lastly, the spin-orbit coupling, prevents the spins from reaching full alignment. This allows for the opening and closing of the superconducting gap. Such phase transitions occur at $E_Z = \sqrt{\Delta_{\text{ind}}^2 + \mu^2}$, where μ is the chemical potential in the semiconductor. The topological regime corresponds to $E_Z > \sqrt{\Delta_{\text{ind}}^2 + \mu^2}$, where Majorana modes can emerge at the ends of one-dimensional systems.

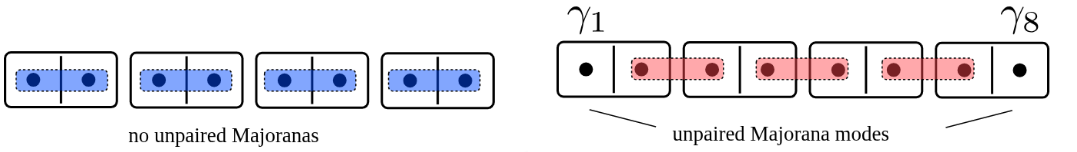

Figure 1: Schematic representation of the Kitaev chain in its two distinct phases. Left: In the trivial phase, Majorana modes on the same site pair to form conventional fermions, leaving no unpaired modes. Right: In the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends, which combine into a non-local fermionic mode. [2]

Another important aspect of hybrid semiconductor-superconductor systems is that applying a magnetic field strong enough to reach the topological regime would break the superconductivity in a pure superconductor. However, in a proximitized semiconductor, the effective g-factor can be much larger than in the superconductor, allowing the Zeeman energy to exceed the induced gap without destroying superconductivity. This, alongside the tunability of the semiconductor's chemical potential via gate voltages, makes these hybrid systems ideal platforms for exploring topological superconductivity and Majorana modes [1].

(ii) Hybrid Systems: Andreev Reflection and Cooper-Pair splitting Microscopically, the proximity effect is a consequence of Andreev reflection, a process where an electron incident on the SM–SC interface is reflected as a hole, while an incoming Cooper pair is absorbed into the superconductor. In confined or low-dimensional hybrid systems, these reflections can lead to discrete subgap states, *Andreev bound states*, that interpolate between conventional superconducting excitations and emergent Majorana modes.
In double-dot systems, this process can become nonlocal: a Cooper pair can effectively split, sending one electron into each region. These nonlocal pairing correlations mimic the nonlocal character of Majorana modes and provide an intuitive picture of how such quasiparticles can emerge from conventional superconducting ingredients.
In summary, hybrid semiconductor–superconductor systems act as the experimental and conceptual link between ordinary s-wave superconductivity and topological superconductivity. The same pairing mechanism that gives rise to Cooper pairs, when combined with spin–orbit coupling and magnetic fields, allows us to engineer effective p-wave pairing, the key ingredient of the Kitaev chain and the Poor Man's Majorana system.

2.4 Engineering Majorana Modes in a Minimal System

2.4.1 Kitaev Chains

We now want to build on the microscopic understanding of superconductivity we investigated in the previous sections. In particular, we want to see how such pairing phenomena can be engineered to host topological excitations, in particular, Majorana bound states. The theoretical foundation for this lies in the Kitaev chain model, a one-dimensional toy model that captures the essential ingredients of topological superconductivity in its simplest form [2].
In the figure above 1, we see a schematic representation of the Kitaev chain in its two distinct phases, the trivial phase on the left and the topological phase on the right.
In the trivial phase, the Majorana modes on the same site pair to form conventional fermions, leaving no unpaired modes. In contrast, in the topological phase, Majoranas on adjacent sites couple, leaving unpaired modes at the chain ends, which combine into a non-local fermionic mode. At first glance, it may seem impossible to isolate a single Majorana mode, since condensed matter systems are composed of electrons, which always correspond to pairs of Majoranas. However, it turns out that by engineering the Hamiltonian appropriately, one can create a

situation where two Majorana modes localize at opposite ends of a one-dimensional system, effectively separating them spatially. This separation is crucial for their non-Abelian statistics and potential applications in topological quantum computing [2].
The Hamiltonian of the Kitaev chain is given by:

$$H = -\mu \sum_{j=1}^{N} \left(c_j^\dagger c_j - \frac{1}{2} \right) - t \sum_{j=1}^{N-1} \left(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j \right) + \Delta \sum_{j=1}^{N-1} \left(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger \right), \tag{3}$$

where μ is the chemical potential, or onsite energy, t is the hopping amplitude, and Δ is the p-wave pairing amplitude. The fermionic operators $c_j^\dagger$ and c_j create and annihilate an electron at site j, respectively.
With the parameters $t = \Delta = 0$ and $\mu \neq 0$, the system is in a trivial insulating phase with no special edge states. However, tuning the parameters such that $t = \Delta > 0$ and $\mu = 0$, the edges become isolated, leaving two unpaired Majorana modes localized at the ends. These modes combine into a single non-local fermionic mode that costs zero energy to occupy. We will later explore how this model maps onto the QD–SC–QD system, and how the parameters can be tuned experimentally to reach this regime.

Bulk Spectrum and Topological phases To understand when the Kitaev chain supports topologically non-trivial states, we now turn from the real-space picture to its bulk description in momentum space. This transformation allows us to analyze the quasiparticle excitation spectrum and identify the conditions under which the superconducting gap closes and reopens. Using translation invariance, we apply a Fourier transform,

$$c_j = \frac{1}{\sqrt{N}} \sum_p e^{ipaj} c_p,$$

This allows us to write the Hamiltonian (3) in momentum space p as:

$$H = \sum_p \xi_p \left(c_p^\dagger c_p - \frac{1}{2} \right) - \sum_p t \cos pa + \Delta \sum_p \left(c_p^\dagger c_{-p}^\dagger e^{ipa} + c_{-p} c_p e^{-ipa} \right), \tag{4}$$

with $\xi_p = -\mu - 2t \cos pa$ and a the lattice spacing. This Hamiltonian becomes diagonal in the Bogoliubov–de Gennes (BdG) formalism. The resulting quasiparticle energy spectrum is:

$$E_{p,\pm} = \pm \sqrt{(\mu + 2t \cos pa)^2 + 4\Delta^2 \sin^2 pa}, \tag{5}$$

The spectrum is gapped for most parameter values, but the gap closes when

$$|\mu| = 2|t|$$

See Figure 2 for a visualization of the quasiparticle energy spectrum. This marks the critical point separating two distinct superconducting phases. When μ crosses this boundry, the system undergoes a topological phase transition: the character of the ground state changes without any symmetry breaking, but through a change in the topology of the quasiparticle band structure [1].

Majorana Representation and Zero Modes In order to further understand the nature of these boundry states, we can express each fermionic operator in terms of two real Majorana operators:

$$c_j = \frac{1}{2}(\gamma_j^A + i\gamma_j^B), \quad c_j^\dagger = \frac{1}{2}(\gamma_j^A - i\gamma_j^B), \tag{6}$$

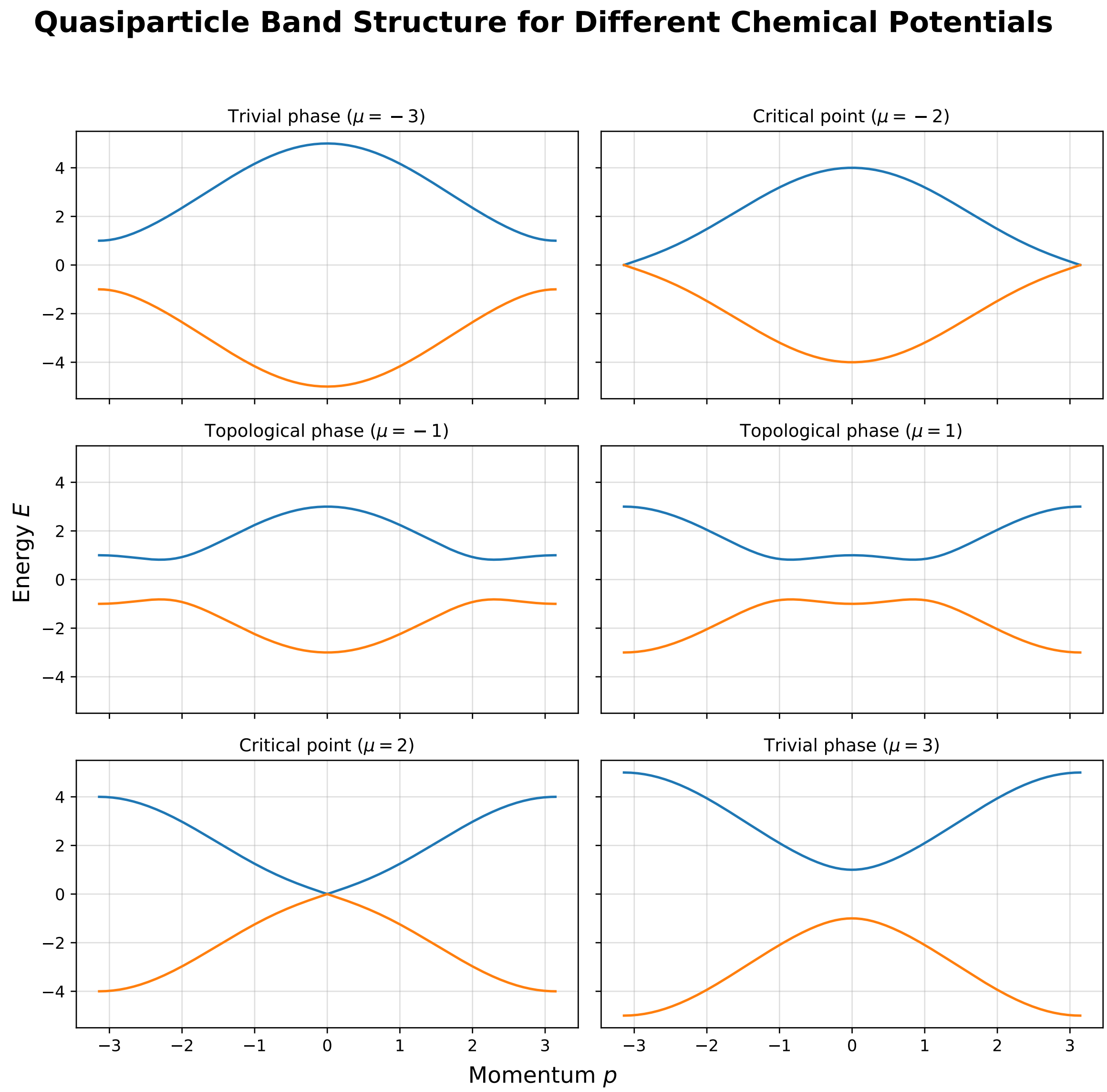

Figure 2: Quasiparticle energy spectrum of the Kitaev chain for different chemical potentials μ. Left: In the trivial phase ($|\mu| > 2|t|$), the spectrum is fully gapped with no zero-energy modes. Right: In the topological phase ($|\mu| < 2|t|$), the gap closes and reopens, signaling the emergence of zero-energy edge modes.

where $\gamma_j^{A,B}$ satisfy $\{\gamma_j^\alpha, \gamma_k^\beta\} = 2\delta_{jk}\delta_{\alpha\beta}$ and $(\gamma_j^\alpha)^\dagger = \gamma_j^\alpha$.
In this basis, the two distinct phases correspond to different pairing patterns between the Majorana operators. For a large μ, ($|\mu| > 2|t|$), the Majorana operators pair up on the same site, leading to a trivial phase with no zero-energy modes. On the other hand, in the topological phase ($|\mu| < 2|t|$) and $t = \Delta$, the Majoranas on adjacent sites couple (γ_j^B with γ_{j+1}^A), leaving two unpaired Majorana modes at the ends of the chain: γ_1^A at the left end and γ_N^B at the right end, see Figure 1.
These two unpaired Majoranas combine into a single non-local fermionic mode:

$$f = \frac{1}{2}(\gamma_1^A + i\gamma_N^B), \tag{7}$$

which satisfies $\{f, f^\dagger\} = 1$ and costs zero energy. The two possible occupations of this non-local fermion correspond to a twofold degenerate ground state. This degeneracy is topologically protected: local perturbations cannot lift it unless they close the bulk gap or couple the two edge Majoranas directly.
The Kitaev chain thus provides the minimal theoretical framework for understanding Majorana zero modes. Its simplicity allows for exact solutions and clear physical intuition, making it an ideal starting point for exploring more complex systems that can host topological superconductivity and Majorana fermions.

2.4.2 The QD-SC-QD System as an Effective Kitaev Chain

While the Kitaev chain provides the minimal theoretical framework for understanding Majorana zero modes, it remains an abstract model, assuming spinless fermions and idealized p-wave pairing. Real materials, however, are spinful and typically host conventional s-wave superconductivity. The challenge, therefore, is to engineer hybrid systems whose effective low-energy behaviour mimics that of the Kitaev chain.
A particularly compact realization of such a system is the double quantum dot coupled via a superconducting segment, the so-called *Poor Man's Majorana* (PMM) setup. This device consists of two quantum dots (QDs) connected through a central superconductor, forming the minimal two-site analogue of the Kitaev chain. Despite its simplicity, it reproduces the essential ingredients of topological superconductivity: coherent hopping, non-local pairing, and the emergence of near-zero-energy states with Majorana-like characteristics.

Mapping to the Kitaev Hamiltonian Each quantum dot represents a site in the minimal Kitaev chain. The Hamiltonian of the hybrid system can be written in a general form as:

$$H = \sum_{i=L,R} \epsilon_i n_i + t(d_L^\dagger d_R + d_R^\dagger d_L) + \Delta(d_L^\dagger d_R^\dagger + d_R d_L) + U \sum_{i=L,R} n_i \tag{8}$$

where $d_{L/R}^\dagger$ and $d_{L/R}$ are electron creation and annihilation operators on the left and right dots. n_i is the number operator for dot i. The parameters $\epsilon_{L,R}$ correspond to the dot energy levels, t is the elastic cotunneling amplitude, Δ is the crossed Andreev reflection amplitude, and U is the on-site Coulomb repulsion energy.
The meaning of each parameter is as follows:

- **Elastic cotunneling (ECT):** An electron tunnels coherently from one dot to the other via a virtual process through the superconductor. This process defines the effective hopping amplitude t, analogous to the kinetic term in the Kitaev model.
- **Crossed Andreev reflection (CAR):** A Cooper pair in the superconductor splits such that one electron tunnels into each dot. This process produces an effective non-local pairing term Δ, which is mathematically equivalent to the p-wave pairing amplitude in the Kitaev chain.

- **On-site energy:** The local dot energies ϵ_L and ϵ_R play the role of the site-dependent chemical potential μ.

At the so-called sweet spot, where $|\epsilon_L| = |\epsilon_R| = 0$ and $|t| = |\Delta|$, the system enters a regime in which two Majorana-like states localize predominantly on separate dots. These two states form an effective non-local fermion and exhibit many of the qualitative features of true topological Majorana bound states, such as zero-energy modes and near-perfect particle–hole symmetry. However, since the system is finite and lacks a bulk, it does not possess true topological protection, hence the name "poor man's Majoranas". [1]

Low-energy Features and Experimental Tunability The poor man's Majorana setup, double quantum dot coupled via a superconductor, provides a minimal realization of the Kitaev chain. In reality, the complete microscopic Hamiltonian includes interactions and spin degrees of freedom. However, in the regime relevant for Poor Man's Majoranas, the low-energy physics is effectively captured by the spinless two-site model introduced above. In this setup, near-zero energy Majorana-like states emerge across the two dots, exhibiting an even- and odd-parity ground state and strong non-local correlations.
This platform is an experimentally attractive method to realize Kitaev chain physics because its parameters can be tuned independently. Gate voltages control the dot energy levels, the transparency of the QD–SC interfaces governs the effective hopping and non-local pairing, and magnetic fields can suppress unwanted spinful excitations. By sweeping these parameters, one can access and identify the Majorana-like properties.
This minimal setup captures the essential low-energy properties of a Kitaev chain while remaining experimentally controllable, providing a direct route from theoretical modeling to realizable quantum devices.

2.4.3 Identifying Majorana Modes in the QD-SC-QD System

In this section we will briefly go through the key signatures that indicate the presence of Majorana-like modes in the QD–SC–QD system. These signatures arise from the unique properties of Majorana fermions. There are a few quantities we can use to identify the presence of PMMs, and their Robustness to perturbations.

The parity Operator The parity operator itself is not a measurement of a Majorana mode, but it is a useful tool in the identification process. The mean-field Hamiltonian of a superconductor does not conserve particle number, but it does conserve fermion parity. What this means is that the number of electrons in the system can change by pairs, but not by single electrons. The parity operator is defined as:

$$P = (-1)^N = \Pi_n(1 - 2c_n^\dagger c_n)$$

where N is the total number of electrons in the system, and $c_n^\dagger$ and c_n are the creation and annihilation operators for the fermionic mode formed by the two Majorana modes. The eigenvalues of the parity operator are $+1$ for even parity (an even number of electrons) and -1 for odd parity (an odd number of electrons).
When working with Majorana modes, we often consider the two degenerate ground states of the system, which differ by their fermion parity. The presence of a Majorana mode implies that these two states are nearly degenerate and can be transformed into each other by adding or removing a single electron, thus changing the parity.

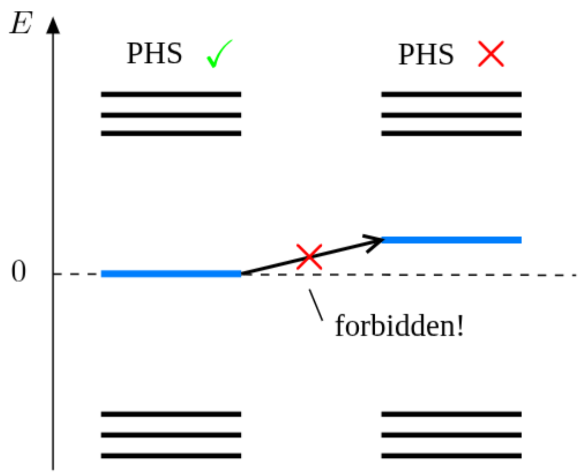

Figure 3: Energy spectrum showing how the energy eigenvalues of the system must be symmetric around zero energy due to particle-hole symmetry [2].

Energy Degeneracy When we try to identify Majorana modes in the QD–SC–QD system, one of the primary signatures we look for is the presence of energy degeneracy (Figure 3) between the even and odd parity ground states. We also want the ground state energies to be separated by a finite energy gap from the excited states. This energy gap protects the Majorana modes from local perturbations and thermal excitations, which is crucial for their stability and potential use in quantum computing. This is because when we want to do quantum computation using Majorana modes, we need to be able to manipulate the states without causing transitions to higher energy states.

Local Distinguishability A property of the Kitaev chain in its topological phase is that measurements done locally cannot reveal if the system is in its odd or even ground state. Meaning the two ground states are locally indistinguishable. In order to quantify this property, we will define the local distinguishability parameter LD as:

$$\mathrm{LD} = \frac{1}{N}\sum_{j=1}^{N} ||\rho_j^o - \rho_j^e|| = \frac{1}{N}\sum_{j=1}^{N} ||\delta\rho_j||$$

[6], [7]. Here, $\rho_j^{o/e}$ are the reduced density matrices of site j for the odd and even parity ground states, respectively. The norm $||\cdot||$ is the trace norm, which measures the difference between the two local states. The LD parameter quantifies how distinguishable the two ground states are when only local measurements are considered. A value of LD close to zero indicates that the two states are nearly indistinguishable locally, which is a hallmark of Majorana modes. Conversely, a value close to one indicates that the states can be easily distinguished by local measurements, suggesting the absence of Majorana modes.

Majorana Polarization In interaction Kitaev chains with finite magnetic fields, there are no ideal Majorana sweet spots, as the presence of the other spin species introduces corrections to the simple picture of Majorana modes. However, sweet spots with very good Majorana localization and close to degenerate ground states with even and odd parity can still appear in the system. To quantify how close we are to an ideal Majorana mode, we can use the Majorana polarization (MP) metric. The MP is a measure of how well the Majorana modes are localized at the ends of the chain and how closely they resemble ideal Majorana fermions. The MP is defined as:

$$M_\alpha = \frac{W_\alpha^2 - Z_\alpha^2}{W_\alpha^2 + Z_\alpha^2}$$

where,

$$W_\alpha = \sum_\sigma \langle o| c_{\alpha\sigma} + c_{\alpha\sigma}^\dagger |e\rangle\,, \quad Z_\alpha = i\sum_\sigma \langle o| c_{\alpha\sigma} - c_{\alpha\sigma}^\dagger |e\rangle$$

Here, α denotes the site index (left or right dot), σ is the spin index, and $|o\rangle$ and $|e\rangle$ are the odd and even parity ground states, respectively. In an ideal sweet spot where two Majoranas are perfectly localized on the left and right dots, the MP values would be $M_L = -M_R = \pm 1$. [1].

Charge Expectation and non-locality The last measurable property we will discuss is the charge expectation value of the ground states. This id measured by taking the expectation values of the number operators in the odd and even ground states:

$$\langle e| n_{L(R)} |e\rangle\,, \quad \langle o| n_{L(R)} |o\rangle$$

where $n_{L(R)}$ is the number operator for the left (right) dot. In the ideal case, these expectation values should be equal for both ground states. This is because, if the difference is zero, the two ground states share the same local charge distribution, making them locally indistinguishable.

2.5 Braiding of Majoranas

The process of braiding Majorana modes, in simple terms, involves exchanging the positions of two Majorana modes in a way that their quantum states are transformed. The reason braiding of Majorana modes is important to discuss is because it is the key operation that allows us to manipulate the quantum information stored in these modes. Braiding refers to the process of exchanging the positions of two Majorana modes in a way that their quantum states are transformed according to non-Abelian statistics. This is a fundamental operation for topological quantum computing, as it enables the implementation of quantum gates that are robust against local perturbations.
Braiding in the right patterns, with the right systems, can lead to the implementation of a universal set of quantum gates **??**, which is necessary for performing arbitrary quantum computations. The non-local nature of Majorana modes means that the information they encode is protected from local sources of decoherence, making them ideal candidates for fault-tolerant quantum computing.SOURCE

2.5.1 Quantum Gates

In quantum computing, and specifically the quantum circuit model of computation, a quantum gate is a basic quantum circuit operating on a small number of qubits. Quantum gates are the building blocks of quantum algorithms, analogous to classical logic gates in classical computing. They manipulate the state of qubits through unitary transformations, allowing for the creation of superposition and entanglement, which are essential for quantum computation.

Pauli gates in terms of fermions The gates in focus for this thesis are the Pauli gates, which are represented by the Pauli matrices:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{9}$$

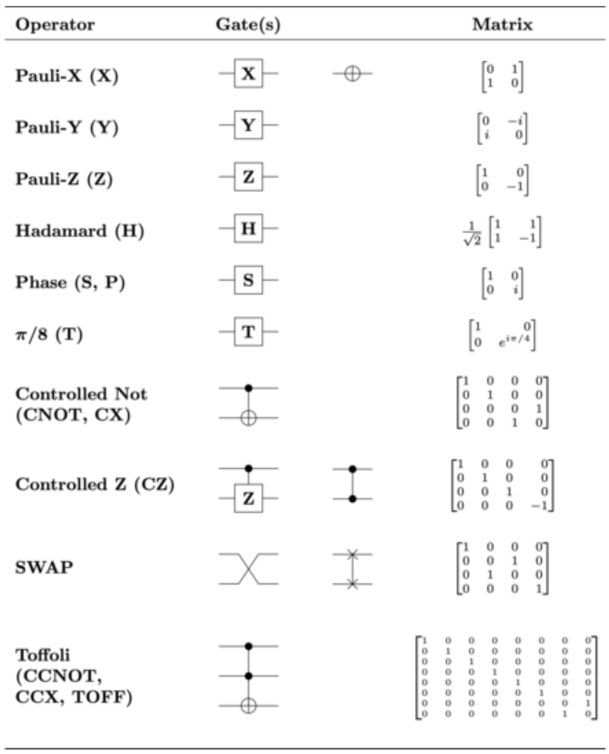

Figure 4: List of common quantum gates and their matrix representations. Wikipedia Quantum logic gates

Suppose you have a pair of quantum states $|0\rangle$ and $|1\rangle$, which can be represented as column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{10}$$

Applying the Pauli gates to these states results in the following transformations:

$$\begin{aligned} X|0\rangle &= |1\rangle, \quad X|1\rangle = |0\rangle \\ Y|0\rangle &= i|1\rangle, \quad Y|1\rangle = -i|0\rangle \\ Z|0\rangle &= |0\rangle, \quad Z|1\rangle = -|1\rangle \end{aligned}$$

The Pauli-X gate, often reffered to as the NOT gate, flips the state of a qubit. The Pauli-Y gate combines a bit flip with a phase flip, while the Pauli-Z gate applies a phase flip to the $|1\rangle$ state. These gates are fundamental for quantum computation and can be implemented through braiding operations in systems hosting Majorana modes. Other quantum gates can be constructed from these basic gates, and together they form a universal set for quantum computation. The ability to perform these operations through braiding of Majorana modes is what makes them so intriguing for topological quantum computing.
In order to find out how to implement these gates through baiding, we must first rewrite these gates in terms of simple fermionic operators.

$$\begin{aligned} c_i^\dagger &\quad \text{creates a fermion at site } i \\ c_i &\quad \text{annihilates a fermion at site } i \end{aligned}$$

For a simple two-site system, we can define the fermionic operators for the two sites as c_1 and c_2. The Pauli gates can be expressed in terms of these operators as follows:

$$
\begin{aligned}
X &= c_1^\dagger c_2 + c_2^\dagger c_1 \\
Y &= -i(c_1^\dagger c_2 - c_2^\dagger c_1) \\
Z &= c_1^\dagger c_1 - c_2^\dagger c_2
\end{aligned}
$$

which quickly can be verified by applying these operators to the basis states $|0\rangle$ and $|1\rangle$.

2.5.2 Jordan-Wigner transforms

The Jordan-Wigner transformation is a transformation that maps spin operators onto fermionic creation and annihilation operators. The reason we want this mapping is that it becomes much easier for us to implement the spin-1/2 Pauli gates in terms of the fermionic operators, which are more natural to implement in our system. We can redefine our fermionic operators in terms of the Pauli matrices as follows:

$$
\begin{aligned}
\sigma_j^+ &= \frac{1}{2}(\sigma_j^x + i\sigma_j^y) \equiv f_j^\dagger \\
\sigma_j^- &= \frac{1}{2}(\sigma_j^x - i\sigma_j^y) \equiv f_j \\
\sigma_j^z &= 2f_j^\dagger f_j - 1
\end{aligned}
$$

Now we have the correct same-site fermionic relations $\{f_j^\dagger, f_j\} = I$. However on different sites, we have $\{f_j^\dagger, f_k\} \neq 0$, and $[\sigma_j^+, \sigma_k^-] = 0$ for $j \neq k$. So spins on different sites commute, while fermions on different sites anticommute. To fix this, we introduce a string of σ^z operators to ensure the correct anticommutation relations. So we define a new set of fermionic operators as follows:

$$
a_j^\dagger = \Pi_{k=1}^{j-1}(-\sigma_k^z) \cdot f_j^\dagger, \quad a_j = \Pi_{k=1}^{j-1}(-\sigma_k^z) \cdot f_j \tag{11}
$$

The transformed spin operators now have the appropriate fermionic canonical anti-commutation relations:

$$
\begin{aligned}
\{a_j^\dagger, a_k\} &= \delta_{jk} \\
\{a_j^\dagger, a_k^\dagger\} &= 0 \\
\{a_j, a_k\} &= 0
\end{aligned}
$$

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Pauli Spin Matrices on Majorana representation If we rewrite the majorana representation from equation 6 in terms of the Jordan-Wigner transformed fermionic operators, we can express the Pauli gates in terms of the Majorana operators. This allows us to see which couplings we need to implement in order to perform the desired quantum gates through braiding of Majorana modes.

In order to do so, we can first define four majorana operators, two for each site, in terms of creation and annihilation operators:

$$
\begin{aligned}
c_1 &= (\gamma_0 + i\gamma_1)/2 \quad c_1^\dagger = (\gamma_0 - i\gamma_1)/2 \\
c_2 &= (\gamma_2 + i\gamma_3)/2 \quad c_2^\dagger = (\gamma_2 - i\gamma_3)/2
\end{aligned}
$$

Then we can express the Pauli gates in terms of these Majorana operators as follows:

$$
\begin{aligned}
X = c_1^\dagger c_2 + c_2^\dagger c_1 &= \frac{\gamma_0 - i\gamma_1}{2}\frac{\gamma_2 + i\gamma_3}{2} + \frac{\gamma_2 - i\gamma_3}{2}\frac{\gamma_0 + i\gamma_1}{2} \\
&= \frac{i}{2}(\gamma_0\gamma_3 - \gamma_1\gamma_2)
\end{aligned}
$$

From the odd parity subspace we have the total Parity operator $P_{tot} = i^2\gamma_0\gamma_1\gamma_2\gamma_3 = -I$. From this we can try to find a relation between $\gamma_0\gamma_3$ and $\gamma_1\gamma_2$.

$$\begin{aligned}
\gamma_0\gamma_1\gamma_2\gamma_3 &= I \\
\gamma_0\gamma_1\gamma_2\gamma_3\gamma_3 &= \gamma_3 \\
\gamma_0\gamma_1\gamma_2 &= \gamma_3 \\
\gamma_1\gamma_2\gamma_0\gamma_0 &= \gamma_3\gamma_0 \\
\gamma_1\gamma_2 &= \gamma_3\gamma_0 = -\gamma_0\gamma_3
\end{aligned}$$

$$\begin{aligned}
Y = -i(c_1^\dagger c_2 - c_2^\dagger c_1) &= -i\left(\frac{\gamma_0 - i\gamma_1}{2}\frac{\gamma_2 + i\gamma_3}{2} - \frac{\gamma_2 - i\gamma_3}{2}\frac{\gamma_0 + i\gamma_1}{2}\right) \\
&= -\frac{i}{2}(\gamma_0\gamma_2 + \gamma_1\gamma_3)Y \qquad = -i\gamma_0\gamma_2 \text{ or } Y = -i\gamma_1\gamma_3
\end{aligned}$$

$$\begin{aligned}
Z = c_1^\dagger c_1 - c_2^\dagger c_2 &= \frac{\gamma_0 - i\gamma_1}{2}\frac{\gamma_0 + i\gamma_1}{2} - \frac{\gamma_2 - i\gamma_3}{2}\frac{\gamma_2 + i\gamma_3}{2} \\
&= \frac{i}{2}(\gamma_0\gamma_1 + \gamma_2\gamma_3)Z \qquad = i\gamma_0\gamma_1 \text{ or } Z = i\gamma_2\gamma_3
\end{aligned}$$

By using the parity conservation and the relation between the Majorana operators, we can express the Pauli gates in terms of different pairs of Majorana operators. So what this means is we can represent the Pauli gates as specific couplings between the Majorana modes:

$$\begin{aligned}
X &= \frac{i}{2}(\gamma_0\gamma_3 - \gamma_1\gamma_2) = i\gamma_0\gamma_3 = -i\gamma_1\gamma_2 \\
Y &= -\frac{i}{2}(\gamma_0\gamma_2 + \gamma_1\gamma_3) = -i\gamma_0\gamma_2 = -i\gamma_1\gamma_3 \\
Z &= \frac{i}{2}(\gamma_0\gamma_1 + \gamma_2\gamma_3) = i\gamma_0\gamma_1 = i\gamma_2\gamma_3
\end{aligned}$$

Majorana Exchange Gates We can take a step towards braiding by defining Majorana exchange operators. We need to find unitaries B_{ij} that exchange Majoranas γ_i and γ_j while leaving the others unchanged. Meaning they should act as follows:

$$\begin{aligned}
B_{ij}\gamma_i B_{ij}^\dagger &= \gamma_j \\
B_{ij}\gamma_j B_{ij}^\dagger &= -\gamma_i \\
B_{ij}\gamma_k B_{ij}^\dagger &= \gamma_k \quad \text{for } k \neq i, j
\end{aligned}$$

So we need to express these unitary gates in terms of the Majorana operators, in a way that relates them to the Pauli gates we want to implement. It turns out that the Majorana exchange operators can be expressed as:

$$B_{ij} = \frac{i}{\sqrt{2}}(1 - \gamma_i\gamma_j) \quad B_{ij}^\dagger = \frac{1-i}{\sqrt{2}}(1 + \gamma_i\gamma_j) \tag{12}$$

Using the combinations of Majorana operators we just found for the Pauli gates, we can express the Majorana exchange operators in terms of the Pauli gates as follows:

$$\begin{aligned}
B_{12}^2 &= -i\gamma_1\gamma_2 = -X \\
B_{03}^2 &= i\gamma_0\gamma_3 = X \\
B_{02}^2 &= i\gamma_0\gamma_2 = -Y \\
B_{13}^2 &= i\gamma_1\gamma_3 = -Y \\
B_{01}^2 &= i\gamma_0\gamma_1 = Z \\
B_{23}^2 &= i\gamma_2\gamma_3 = -Z
\end{aligned}$$

2.5.3 Kato Berry Phase

Berry phase is a geometric phase acquired by the wavefunction of a quantum system when it undergoes adiabatic evolution around a closed loop in parameter space. In the context of Majorana modes, the Berry phase can be used to characterize the topological properties of the system and to understand how the quantum states transform under braiding operations.
In this thesis, the berry phase will help us measure wether or not we manage to implement the desired quantum gates through braiding of Majorana modes. By calculating the Berry phase associated with the braiding process, we can determine if the resulting transformation corresponds to the expected unitary operation on the qubit space defined by the Majorana modes.

If we follow a non-degenerate eigenstate of a Hamiltonian as we slowly (adiabatically) change a parameter in a loop in parameter space, we will end up with the same state up to a phase. The phase has two contributions, one is dynamical and depends on the actual time scale of the evolution and the energy of the state. But there is another contribution that is independent of the details of the evolution. It only depends on the path taken. It is a geometric quantity and comes from a sort of curvature in the space of eigenstates. It is called the Berry phase and can be calculated from the Berry connection $A = i \langle d\psi|\psi|d\psi|\psi\rangle$ integrated during the evolution. If the state is degenerate, we can get more than a phase. Even for adiabatic evolution, the degenerate states will mix together. Thus, the Berry phase will be represented by a matrix that mixes the states together. The Berry connection is also a matrix with a dimension equal to the number of degenerate states: $A_{nm} = i \langle d\psi_n|\psi_m|d\psi_n|\psi_m\rangle$.

A nice way to calculate non-abelian Berry phase is known as the Kato method. This method uses the projector P that projects on the two degenerate ground states. Then one solves the differential equation

$$\partial_t U_t = i\,[P, \partial_t P]\, U_t U_0 = I$$

and the final unitary is the non-abelian Berry phase U_T. It shouldn't matter if the total time T is long or short. UT is the gate that a proper adiabatic evolution would implement. Note that the projector P is time-dependent. It can be calculated numerrically by diagonalizing the Hamiltonian, or analytically if the hamitonian is simple enough.

2.5.4 Effective Braiding Hamiltonian

If we assume that we already have a system with four Majorana modes, we are able to write down an effective Hamiltonian that describes the couplings between these Majorana modes. In the figure above 5, we see a schematic representation of a system hosting six Majorana modes, arranged in a Y-junction. The Majorana modes are represented by the X-markers. In the figure we see that the Majoranas are coupled in pairs. This represents a standard initial state of three Poor Man's Majorana setups, where each pair represents the emerging edge modes of each setup. The four most relevant Majorana modes for our braiding operation are the ones labeled as γ_0, γ_1, γ_2, and γ_3. While the other two Majoranas, γ_4 and γ_5, are coupled together and form a non-local fermionic mode that is not involved in the braiding process. The effective Hamiltonian describing the couplings between the four Majorana modes can take the simple form of:

$$\begin{aligned} H =& \Delta_{01} i\gamma_0\gamma_1 + \\ & \Delta_{02} i\gamma_0\gamma_2 + \\ & \Delta_{03} i\gamma_0\gamma_3 \end{aligned} \tag{13}$$

where Δ_{ij} represents a tunable coupling strength between Majorana modes γ_i and γ_j. In reality this Δ_{ij} will be a function of the physical parameters of the system, such as the tunneling amplitudes, the superconducting gap, and the gate voltages. By tuning these parameters to the

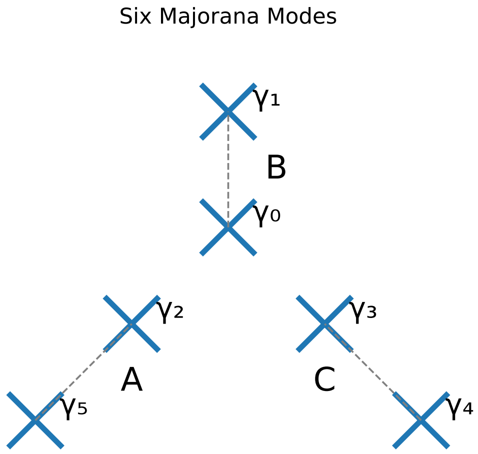

Figure 5: Schematic representation of a system hosting six majoranas. The illustrations shows pairs of two and two Majoranas coupling together, in a Y-junction, each pair form a subsystem that can be described by a Poor Man's Majorana setup, A, B and C.

correct values, we can control the couplings between the Majorana modes and thus implement the desired braiding operations.

The Δ in the effective Hamiltonian is a simplified version of what we would have in a real system. In a more realistic model, we would replace these, with a intra PMM junction coupling. However, if we look back at our Poor Mans Hamiltonian 8, the recipe for the junction Hamiltonian are the same coupling effects as the effective Hamiltonian. In essence, we are just extending a Poor Man's Majorana to where if a coupling is turned on, we just have one longer PMM chain, and one shorter. So if we want to couple γ_0 and γ_2, we would now have a chain connecting $\gamma_0, \gamma_1, \gamma_2, \gamma_5$. Where the coupling now can be represented by turning on a new superconducting segment between the two PMM setups. This junction Hamiltonian would simply be:

$$H_{JuncAB} = t_{AB}(d_A^\dagger d_B + d_B^\dagger d_A) + \Delta_{AB}(d_A^\dagger d_B^\dagger + d_B d_A) \tag{14}$$

but if we are simply testing out if our braiding protocol works, and we have stable Majorana modes, we can just use the effective Hamiltonian in equation 13 to simulate the braiding process and calculate the resulting Berry phase. This allows us to verify if the braiding operation implements the desired quantum gate on the qubit space defined by the Majorana modes.

2.6 Sweet spot parameter search

The Poor Man's Majorana setup is a highly tunable system, with several parameters that can be adjusted to access different regimes of the system. We know of certain non-interacting sweet spots where the Majorana modes are perfectly localized and the ground states are exactly degenerate. However, in the presence of interactions, these ideal sweet spots may no longer exist.
In an ideal system, we would have a perfect sweet spot where the Majorana modes are perfectly localized on the left and right dots and with degeneracy protection accross multiple energy levels, for topological protection.

As we found in 2.4.1, we very well know the ideal parameters for a non-interacting two-dot system. However, when introducing interactions, the parameter space becomes slightly more complex. And it does not seem possible to find degeneracy accross multiple energy levels, which

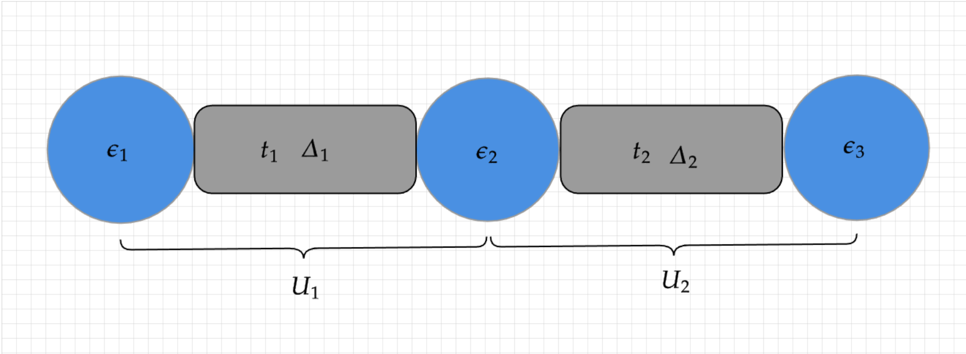

Figure 6: Schematic representation of the PMM setup. The three quantum dots are represented by the blue circles, and the superconducting segments are represented by the grey rectangles. The tunable parameters of the system, such as the dot energy levels, ECT, CAR and the coloumb interaction, are indicated by their respective labels. By tuning these parameters, we can access different regimes of the system and identify the sweet spots where Majorana-like modes emerge.

is preferable for topological protection. However, there is no evidence that it is not possible to find such a sweet spot if you increase the number of quantum dots in the system, creating longer chain. Figure 6 shows a schematic representation of a three-dot system, where we can see that there are multiple tunable parameters. The more dots we add, the more tunable parameters we have. But with more tunable parameters, the problem of finding a sweet spot analytically becomes very complex, especially if we dont rely on symmetries and assumptions. Therefore, we will need to explore the parameter space numerically, using optimization techniques to find the sweet spots where Majorana-like modes emerge.

2.6.1 General Loss function

For a simple two-dot system, we have a total of 5 tunable parameters: the dot energy levels ϵ_L and ϵ_R, the elastic cotunneling amplitude t, the crossed Andreev reflection amplitude Δ, and the on-site Coulomb repulsion energy U. If we introduce a third dot, we get 9 independent tunable parameters, assuming the interaction strengths can be different for each dot. With more dots, the number of tunable parameters increases significantly, making the parameter space more complex and harder to explore analytically, vizualised in 7.

Before setting up the optimization problem, we need to define a loss function that quantifies how close we are to finding a sweet spot with Majorana-like modes. This loss function should take into account the key signatures of Majorana modes, such as energy degeneracy and gap, local indistinguishability, Majorana polarization, and charge expectation values. By minimizing this loss function, we can identify the regions in parameter space where Majorana-like modes are likely to emerge.

The loss function can be defined as a weighted sum of the different metrics that characterize the presence of Majorana modes. The weights can vary depending on the importance we assign to each metric. For example ground state degeneracy and gap is much more important than the gap and degeneracy of the higher energy levels, so we can assign a higher weight to the first one, gradially decreasing the weights for the higher energy levels. Other metrics, defining for Majoranas, such as local indistinguishability, charge expectation values, and Majorana polarization should be included with a significant weight, as they are crucial for the identification of Majorana modes.

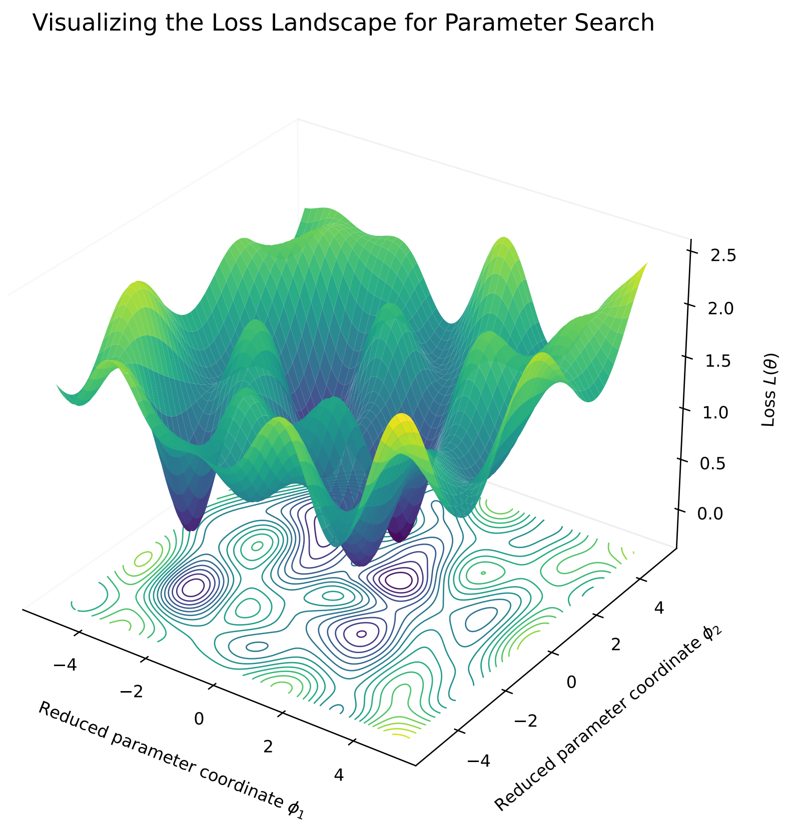

Figure 7: Visualization of the parameter space for an n-dot system, showing how multiple parameters and their combinations create a complex landscape with multiple local minima.

2.6.2 Optimization landscape

The search for interacting sweet spots is complicated by the fact that the loss function generally contains many local minima. A purely local optimization routine can therefore converge to a parameter set that is only locally optimal, even though other regions of parameter space may yield better Majorana-like behavior. For this reason, the numerical procedure used in this thesis combines local gradient-based minimization with repeated restarts and basin-hopping. The detailed implementation is described in Section 3.1.3, while Figure 7 serves as a schematic illustration of the complex parameter landscape that motivates this approach.

3 Method

This chapter describes the methods used to optimize the Majorana modes in the junction setup and to analyze their properties. We first discuss the parameter optimization process, including the model parameterization, loss function design based on Majorana criteria, and the optimization strategy. Next, we detail the post-optimization analysis of the setup, covering Hamiltonian reconstruction, parity classification, and various diagnostics. Finally, we describe the braiding protocol implemented in an effective Majorana model and how we validate the results using gate-validation checks.

3.1 Parameter Optimization

A key part of this thesis is the numerical search for parameter regimes in which an interacting n-dot PMM system exhibits Majorana-like low-energy states. For the non-interacting two-dot problem, the sweet-spot conditions are known analytically, but once Coulomb interactions, longer chains, and constraints on higher excited states are included, the problem quickly becomes too complicated to solve analytically. We therefore treat the search for suitable parameters as an optimization problem, where the Hamiltonian is evaluated repeatedly and compared against a set of Majorana criteria.

The purpose of the optimization is not to claim a unique exact sweet spot for a given device. Instead, it is to identify parameter regions that repeatedly produce near-degenerate, well-localized, and spectrally protected Majorana-like states, and to study how these regions change with the number of dots and with the interaction strength. In the calculations presented here, the Coulomb repulsion U is fixed to selected values, while the remaining free parameters are optimized within that interaction regime.

3.1.1 Model parameterization

The model parameterization is simply based on the n-dot PMM Hamiltonian:

$$H = \sum_{i=1}^{n} \epsilon_i n_i + \sum_{i=1}^{n-1} \left[-t_i \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) + \Delta_i \left(c_i c_{i+1} + c_{i+1}^\dagger c_i^\dagger \right) + U n_i n_{i+1} \right] \tag{15}$$

For an n-dot system, this Hamiltonian contains n on-site energies ϵ_i, $n-1$ hopping amplitudes t_i, $n-1$ pairing amplitudes Δ_i, and nearest-neighbor Coulomb terms. The optimization is carried out on a reduced parameter vector containing only the free parameters. Depending on the chosen ansatz, a parameter can either be treated as homogeneous, inhomogeneous, or fixed to a prescribed value. This makes it possible to compare different levels of model complexity while keeping the numerical procedure the same.

In the calculations used throughout this thesis, the starting point is chosen close to the known non-interacting sweet spot, with $t_i \approx \Delta_i$ and on-site energies near the corresponding sweet-spot values. The interaction strength U is then fixed, and the remaining parameters are allowed to adjust so that the optimizer searches for the best Majorana-like configuration within that interaction regime.

3.1.2 Loss function and Majorana criteria

The loss function is designed to optimize the parameters of the n-dot PMM Hamiltonian such that the resulting system exhibits properties consistent with Majorana modes. The loss function is based on several criteria that are commonly associated with stable and protected Majorana modes:

Energy degeneracy: Energy degeneracy is highly weighted in the loss function, especially the degeneracy of the ground state. The weight of the energy degeneracy is based on which energy level we are looking at. For example, the degeneracy of the ground state is more important than the degeneracy of the first excited state, and so on. This is because the ground-state degeneracy is a key signature of Majorana modes, while degeneracies in higher energy levels may not be as robust or relevant for topological protection. The array of weights linearly decreases from some weight ω_{max} for the ground-state degeneracy to a smaller weight ω_{min} for the degeneracy of the highest energy level considered in the loss function. This way, we can prioritize the optimization towards achieving a degenerate ground state while still considering the overall energy spectrum.
Degeneracy Gaps: The loss function also includes terms that penalize small energy gaps between energy levels. This is not as highly weighted as the energy degeneracy, but it is still important to ensure that there are sufficiently large energy gaps to protect the Majorana modes from thermal excitations and other perturbations. The weighting of the energy gaps is based on the same idea as the energy degeneracy, where the weight linearly decreases from some weight ω_{max} for the gap between the ground state and the first excited state, to a smaller weight ω_{min} for the gap between the highest energy levels considered in the loss function. We want clearly separated energy levels while performing a braiding protocol. Better energy gaps help to ensure that the system remains in the desired energy subspace during the braiding process, which is crucial for maintaining the topological protection of the Majorana modes.
Charge difference: The function also includes a term that penalizes charge differences between the degenerate states. This is because, when Majorana modes are present, the degenerate states should have the same charge. A significant charge difference between the degenerate states would indicate that they are not true Majorana modes.
Majorana Polarization: On the edge dots of the PMM system, the Majorana polarization should equal ± 1, while the inner dots should always be zero. The loss function calculates the Majorana polarization for each dot and includes terms that penalize deviations from these ideal values. This criterion is important because it reflects the spatial localization of the Majorana modes, which is a key feature for their stability and potential use in quantum computing applications. By optimizing the parameters to achieve the desired Majorana polarization profile, we can enhance the likelihood of realizing robust Majorana modes in the system.

If all of the above criteria are satisfied, the system is likely to host Majorana modes with desirable properties. In practice, the different penalty terms are combined into a single weighted loss and normalized by the mean energy scale of the spectrum, so that different parameter sets can be compared on a more equal footing.

3.1.3 Optimization strategy

The numerical search is carried out in two stages. First, a local gradient-based optimization is performed using the Adam algorithm. The optimizer acts on the reduced parameter vector, while after each update the vector is mapped back to the physical parameter set. In this step, fixed parameters are reinserted and constraints such as positive couplings are enforced. The local optimization therefore explores only physically allowed Hamiltonians, while still benefiting from automatic differentiation of the loss function.

Because the loss landscape contains many local minima, a single local run is usually not sufficient. We therefore supplement the local optimizer with repeated restarts and a basin-hopping procedure. Starting from a trial parameter set, several locally optimized runs are performed from randomly perturbed initial conditions in its neighborhood, and the lowest-loss result is retained as the representative of that basin. This improves the robustness of the search and reduces the risk that the final result depends too strongly on a single initialization.

To move between basins, a random step is added to the current parameter vector before the local refinement is repeated. The new candidate is accepted whenever it improves the loss, and can also be accepted with a Metropolis probability when it is slightly worse. In this way, the search can occasionally leave a shallow local minimum and continue exploring other regions of parameter space. For each choice of n and fixed U, the parameter set with the lowest final loss is kept for the subsequent spectral analysis and braiding calculations.

3.2 Post-Optimization Analysis of the Setup

3.2.1 Hamiltonian reconstruction and parity classification

Once an optimized parameter set has been found, the raw optimization vector is converted back into physical parameters and the full many-body Hamiltonian is reconstructed. This step is important because the optimizer itself only works with a reduced parameter vector, whereas the physical interpretation requires the complete set of hopping, pairing, on-site-energy, and interaction parameters. In practice, fixed parameters are reinserted, homogeneous parameters are expanded where needed, and the Hamiltonian is rebuilt in the same basis used during optimization. For selected configurations, the parameter values can also be exported and reloaded in a separate analysis setup, where the Hamiltonian is reconstructed numerically and, when useful, symbolically.

The reconstructed Hamiltonian is then diagonalized to obtain its full spectrum and eigenstates. To identify the low-energy structure relevant for Majorana physics, the eigenstates are classified according to the total fermion parity operator,

$$P_{\text{tot}} = \prod_{i=1}^{n} (1 - 2n_i).$$

This separates the spectrum into even- and odd-parity sectors, which makes it possible to compare candidate degenerate partners level by level. The parity-resolved spectrum is used to inspect the ground-state splitting, the excitation gap above the low-energy manifold, and the overall pairing of even and odd states. This classification step also defines the low-energy basis that is used later when extracting effective Majorana operators and constructing the braiding protocol.

3.2.2 Spectral and local diagnostics

The optimization condenses several physical criteria into a single scalar loss, but that loss alone does not explain why a parameter set is good. For this reason, each promising configuration is analyzed using a set of complementary diagnostics. At the spectral level, we inspect the parity-resolved energy levels, the total even-odd splitting, and the charge difference between candidate degenerate partners. In addition, the optimized parameters are visualized together with the spectrum to relate spectral features directly to the physical setup.

At the operator level, we compute observables that probe how Majorana-like the low-energy states actually are. These include the Majorana polarization profile across the chain, the local electron occupancy in even and odd states, the matrix elements of local Majorana operators between parity partners, and the hopping and pairing expectation values on each bond. In a separate analysis pass, we also evaluate transition matrices

$$T^{(O)}_{ss'} = \langle s|O|s'\rangle$$

for relevant operators such as local charge operators, local Majorana operators, and nearest-neighbor hopping and pairing terms. These diagnostics help distinguish configurations that merely achieve a low numerical loss from configurations that exhibit a clear low-energy Majorana interpretation. Only after this analysis step do we select configurations for the braiding calculations described in the following sections.

3.3 Braiding in an Effective Majorana Model

3.3.1 Pulse protocol

Before applying the braiding procedure to the optimized microscopic setup, we first test it in an idealized effective model containing four Majorana operators γ_0, γ_1, γ_2, and γ_3. The purpose of this simplified model is to isolate the braiding protocol itself from the complications of the full interacting Hamiltonian. In this setting, the time-dependent Hamiltonian is written as

$$H(t) = \Delta_1(t)\, i\gamma_0\gamma_1 + \Delta_2(t)\, i\gamma_0\gamma_2 + \Delta_3(t)\, i\gamma_0\gamma_3,$$

where the couplings $\Delta_1(t)$, $\Delta_2(t)$, and $\Delta_3(t)$ are turned on and off sequentially through smooth pulses. The pulses are chosen as broadened step-like functions with controllable width and steepness, so that the exchange path remains continuous in parameter space.

The braiding cycle is implemented by keeping one coupling active at the beginning and end of the protocol, while the two remaining couplings are activated at intermediate times. In this way, the dominant Majorana pairing is moved through the network in a prescribed order. At each time step, the instantaneous Hamiltonian is diagonalized and its spectrum is monitored to verify that the ground-state manifold remains approximately degenerate and separated from higher excited states throughout the cycle.

3.3.2 Kato Berry-phase calculation

To determine which gate is implemented by the braiding cycle, we compute the non-Abelian Berry phase of the instantaneous low-energy subspace using the Kato formulation of adiabatic transport. Starting from the degenerate ground-state manifold at $t = 0$, we construct the projector

$$P(t) = \sum_{a\in \mathrm{GS}} |\psi_a(t)\rangle \langle\psi_a(t)|\,,$$

and numerically propagate the associated Kato evolution operator along the discretized path in time. In practice, this is done by diagonalizing the Hamiltonian at each time step, constructing the projector onto the low-energy manifold, and updating the transport operator from the change in the projector between successive steps.

This procedure yields a unitary operator acting on the degenerate subspace at the end of the loop. The resulting unitary can then be compared with the expected Majorana exchange operator. In the effective model, this provides a clean reference for what the ideal braid should do before the same protocol is applied to a more realistic projected Hamiltonian.

3.4 Braiding with Optimized Majorana Modes

3.4.1 Extraction of Majorana operators

After identifying optimized parameter sets, we construct effective Majorana operators directly from the low-energy even- and odd-parity eigenstates of each subsystem. For each subsystem, the even and odd states are paired according to their energy proximity, and operator combinations of the form

$$\gamma_1 \sim |o\rangle \langle e| + |e\rangle \langle o|\,, \qquad \gamma_2 \sim i(|o\rangle \langle e| - |e\rangle \langle o|)$$

are formed. By summing over a chosen number of low-energy parity pairs, we obtain effective Majorana operators that act within the relevant low-energy sector.

These subsystem operators are then embedded into the full multi-arm Hilbert space using Jordan-Wigner strings, so that the resulting Majorana operators satisfy the correct fermionic anticommutation structure across the entire device. In this way, the idealized Majorana degrees of freedom used in the toy model are replaced by operators extracted from the optimized microscopic Hamiltonian.

3.4.2 Projected microscopic junction Hamiltonian

To move beyond the idealized braid Hamiltonian, we construct junction operators that couple neighboring subsystems through microscopic hopping and pairing terms. These operators are first defined in the full Hilbert space and are then projected onto the low-energy manifold spanned by the relevant optimized states. The projected operators,

$$T_{AB} = P^{\dagger} H_{AB}^{\text{junc}} P, \qquad T_{AC} = P^{\dagger} H_{AC}^{\text{junc}} P, \tag{16}$$

replace the hand-chosen Majorana bilinears of the toy model.

The time-dependent braiding Hamiltonian is then built by driving these projected junction operators with the same pulse sequence used in the effective model. This keeps the geometry of the braid fixed while allowing us to test whether the microscopic optimized setup reproduces the desired exchange dynamics in its low-energy sector.

3.4.3 Gate-validation checks

The final step is to test whether the projected microscopic braid behaves like the intended Majorana exchange. Several consistency checks are performed along the braiding path. First, we verify basic structural properties such as Hermiticity, parity conservation, ground-state splitting, and the excitation gap above the low-energy manifold. Next, we test the algebra of the extracted Majorana operators by checking Hermiticity, normalization, and mutual anticommutation.

After the Kato evolution has been computed, the resulting unitary is compared with the expected exchange action on the Majorana operators. In particular, we test whether a single exchange maps $\gamma_2 \to -\gamma_3$ and $\gamma_3 \to \gamma_2$, whether a double exchange produces the expected sign change, and whether repeated exchanges return the system to its initial operator structure. We also project the braid unitary into parity sectors and compare the resulting blocks with the target gate. Together, these checks determine whether the optimized setup supports a braid that is not only spectrally well-defined, but also implements the intended topological operation.

4 Results

This section presents the main numerical results of the thesis. I first report the outcome of the parameter optimization, then analyze the spectral and local signatures of the optimized configurations, and finally study whether the resulting low-energy states support the intended braiding behavior. The broader interpretation of these findings and their main limitations are deferred to the discussion section.

4.1 Parameter Optimization Results

To obtain an overview of the optimization landscape, I first compare the lowest-loss configuration found for each combination of system size n and fixed interaction strength U. This separates the global trends of the optimization procedure from the more detailed diagnostics of individual configurations. For the two-dot system, the optimizer reproduces the known non-interacting sweet spot at $U = 0$, while increasing interaction strength gradually shifts the optimal on-site energies and pairing amplitudes away from that limit. For three and four dots, the optimization landscape becomes more complex, and the best solutions show stronger inhomogeneity in both the onsite energies and the pairing amplitudes.

4.1.1 Dependence on interaction strength and system size

Table 1 summarizes the lowest-loss configuration found for each combination of system size n and fixed interaction strength U. Since the hopping amplitudes were fixed to $t_i = 1$ in these runs, only the optimized pairing amplitudes and on-site energies are listed. Empty entries indicate parameters that do not exist for that system size.

n	U	Loss	Δ_1	Δ_2	Δ_3	ϵ_1	ϵ_2	ϵ_3	ϵ_4
2	0.0	5.657e-03	1.000	–	–	-0.001	0.002	–	–
2	0.1	2.617e-01	1.047	–	–	-0.050	-0.047	–	–
2	0.5	1.083e+00	1.249	–	–	-0.255	-0.249	–	–
2	1.0	1.940e+00	1.498	–	–	-0.499	-0.496	–	–
2	2.0	3.203e+00	2.000	–	–	-1.006	-1.004	–	–
2	5.0	4.004e+00	3.499	–	–	-2.508	-2.506	–	–
3	0.0	3.442e-01	1.010	1.348	–	0.019	1.774	-0.466	–
3	0.1	5.603e-01	1.001	1.002	–	-0.101	-0.607	-0.003	–
3	0.5	9.359e-01	1.004	1.012	–	-0.046	-0.889	-0.454	–
3	1.0	1.452e+00	1.025	1.130	–	-0.195	-1.530	-0.802	–
3	2.0	2.326e+00	1.584	0.978	–	-1.511	-3.077	-0.484	–
3	5.0	3.142e+00	0.846	1.746	–	-4.814	1.009	-0.184	–
4	0.0	3.161e-01	2.698	5.332	1.004	-0.059	-4.252	0.236	-0.003
4	0.1	4.840e-01	1.790	5.002	1.013	0.156	2.376	-1.966	-0.050
4	0.5	1.006e+00	1.212	3.490	1.206	-0.246	-0.499	-0.503	-0.250
4	1.0	5.670e-01	2.238	18.807	1.072	-0.500	-1.028	-1.025	-0.496
4	2.0	1.824e+00	1.492	5.537	1.588	-0.869	3.526	-6.508	-1.220
4	5.0	2.600e+00	3.389	9.513	3.336	-2.645	5.032	-15.431	-2.293

Table 1: Lowest-loss optimized configurations extracted from `configuration.json` for each combination of system size n and fixed interaction strength U.

The reason for a fixed $t_i = 1$ is that if we do not lock the hopping amplitudes, the optimizer can simply increase t_i in order to replicate a non-interacting sweet spot even at strong interactions, which is not the intended behavior. By fixing t_i, we force the optimizer to find nontrivial parameter profiles that satisfy the Majorana-based loss criteria without relying on trivial rescaling. Several trends are immediately visible. For the two-dot system, the optimizer reproduces the expected non-interacting sweet spot at $U = 0$, where $\Delta_1 \approx t_1$ and the on-site energies remain very close to zero. As the interaction strength is increased, the loss grows steadily, the optimized pairing amplitude increases above the fixed hopping, and the on-site energies move approximately toward $\epsilon_i \approx -U/2$. This indicates that the optimizer is deforming the well-known two-dot sweet spot in a smooth and physically interpretable way.

When the interaction strength is increased further, the loss continues to grow, and the optimized parameters deviate more strongly from the non-interacting limit. This can be due to the fact that the system would need stronger hopping and pairing amplitudes to maintain the Majorana-like low-energy structure in the presence of stronger interactions, but since t_i is fixed, the optimizer can only increase Δ_i and adjust ϵ_i to partially compensate. The structure of the loss function could also suggest that it favors lower parameter values, because when the parameters become too large, the energy scales of the system increase which could lead to numerically larger losses, even though the Majorana-like physics might still be present. This is a subtle point that could be clarified by analyzing the loss landscape in more detail, but the overall trend is that stronger interactions push the system away from the simple non-interacting sweet spot and require more significant parameter adjustments to maintain low loss.

4.1.2 Trends in the optimized parameters

For all three system sizes in 1, the best configurations are found where the interaction strength is relatively weak. For all three cases the best solution is found at $U = 0$, which is not surprising since we have well known non-interacting sweet spots that the optimizer can easily find.

For the two-dot system, the optimized parameters at $U = 0$ are very close to the ideal non-interacting sweet spot, with $\Delta_1 \approx t_1 = 1$ and $\epsilon_i \approx 0$. This is also the analytically known solution for the two-dot system, so it serves as a sanity check that the optimizer can reproduce this limit, which is a good indicator that the optimization procedure is working correctly. With a small interaction of $U = 0.1$, the optimized parameters deviate slightly from the non-interacting sweet spot, with Δ_1 increasing to about 1.05. Figure 8 and 9 show how for when increasing the interaction strength to $U = 0.1$ and $U = 2.0$, the parity-resolved spectrum changes, and the excited state becomes more and more split for higher interaction strengths, which is consistent with the fact that the two-dot sweet spot only guarantees degeneracy of the ground states, and not of the excited states.

For the three-dot system, the best configurations at weak and intermediate interaction strengths still remain relatively close to the homogeneous limit, with $\Delta_1 \approx \Delta_2 \approx 1$ for $U = 0.1$ and $U = 0.5$. At the same time, the middle-site energy becomes substantially more negative than the edge-site energies, suggesting that the optimization favors a differentiated central region even when the pairing amplitudes remain nearly uniform. At larger interaction strengths, both the loss and the parameter asymmetry increase, indicating that the system is moving away from the simplest weak-coupling Majorana picture. In figure 10 we can see the parity resolved spectrum and the Majorana polarization profile for the three-dot system at $U = 0.1$. We can see that the parameter optimizer found a set of parameters where all 4 energy levels are degenerate, even when the system is interacting. Looking at the plot showing the Majorana polarization, we can see that the three dots have the expected MP values, where the left quantum dot has a MP close to +1, the right quantum dot has a MP close to -1, and the middle dot has a MP close to zero.

The four-dot system shows the clearest sign that the optimization landscape becomes more

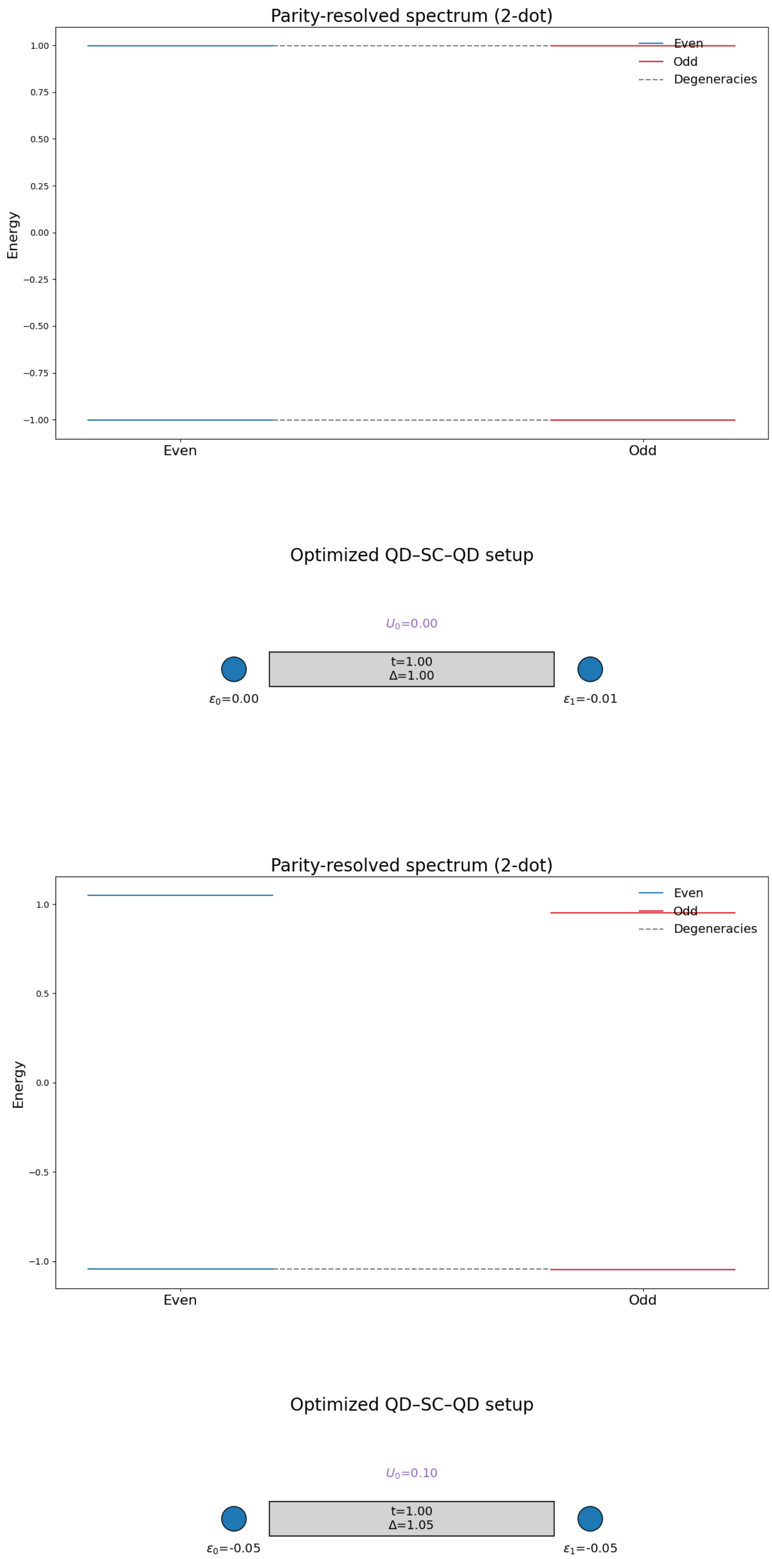

Figure 8: The two plots show the parity resolved sprectrum after optimization for the two-dot system at $U = 0$ and $U = 0.1$ respectively. The optimized parameters are shown schematically below each plot. We can see that for $U = 0$, both energy levels are exactly degenerate, while for $U = 0.1$ the degeneracy of the excited state is slightly lifted.

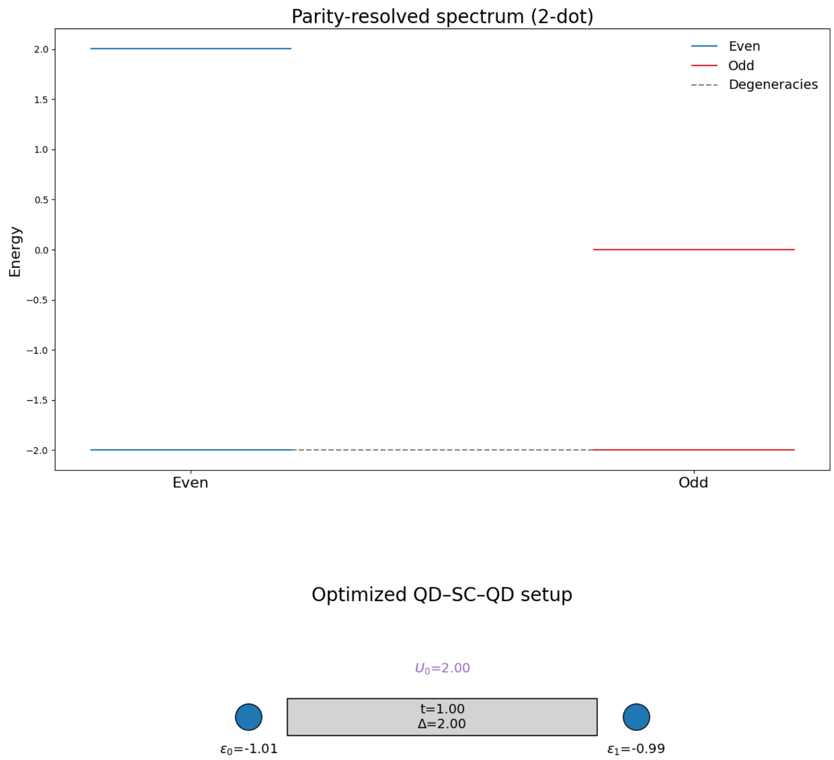

Figure 9: The plot shows the parity resolved sprectrum after optimization for the two-dot system at $U = 2.0$. The optimized parameters are shown schematically below the plot. We can see that for $U = 2.0$, the degeneracy of both the ground state and the excited state is lifted

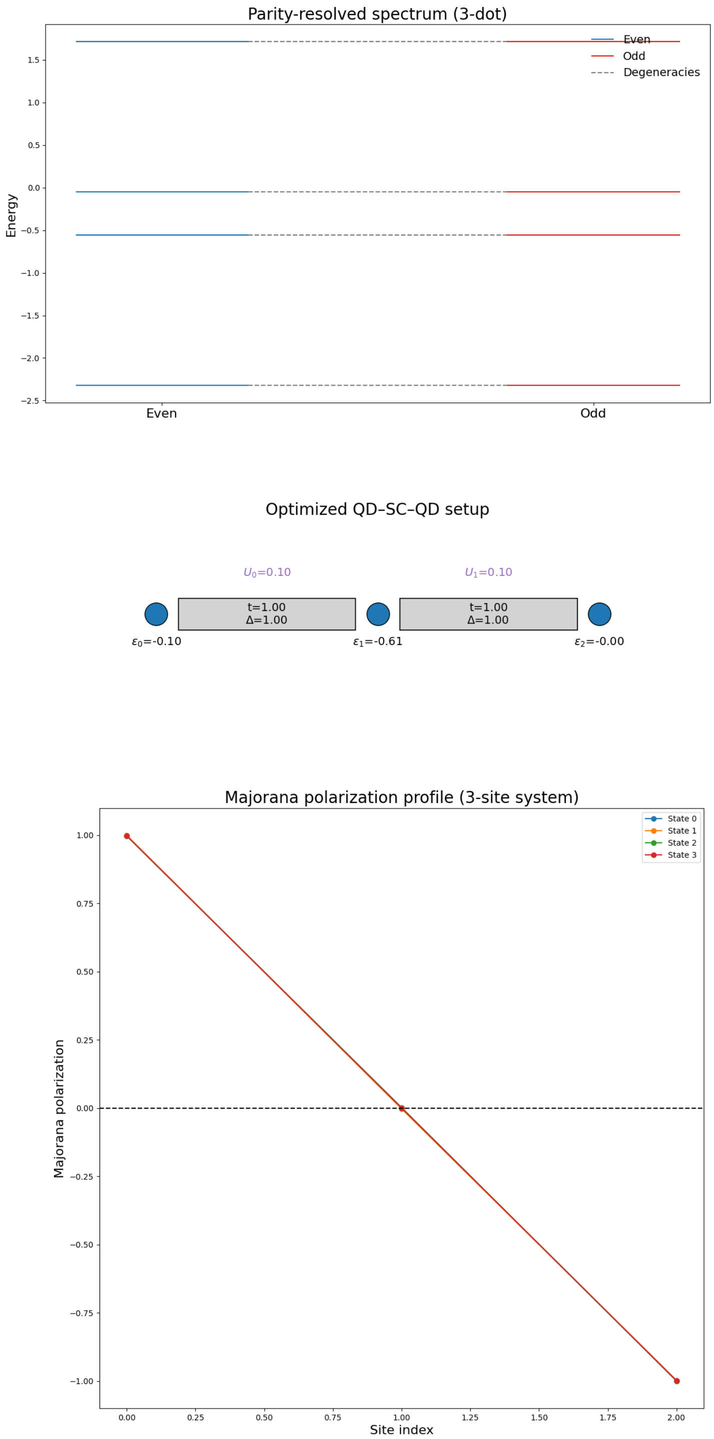

Figure 10: The two plots show the parity resolved sprectrum after optimization for the three dot system, with interaction strength $U = 0.1$. The optimized parameters are shown schematically below the left plot, and the Majorana polarization profile of the lowest-energy states is shown in the right plot.

complicated as the chain length increases. In several of the best configurations, the optimizer strongly enhances the central pairing amplitudes while leaving the outer pairing amplitudes much smaller. This is especially visible for $U = 0$, $U = 0.1$, and $U = 1.0$, where the middle bond is favored much more strongly than the edge bonds. The optimized on-site energies also become highly inhomogeneous, particularly at stronger interactions. Taken together, these results suggest that for longer PMM chains the optimizer does not simply preserve the two-dot sweet-spot structure, but instead searches for more nontrivial inhomogeneous parameter profiles that still satisfy the Majorana-based loss criteria.

An interesting repeated occurance in the optimized parameters is that the optimizer tends to favor larger chemical potentials for the middle sites, $\epsilon_m \gg \epsilon_e$, where m and e refer to middle and edge sites respectively. This is especially visible for the three-dot system, where the middle-site energy becomes substantially more negative than the edge-site energies even at weak interactions 1. Meaning in order to create a Majorana-like structure at the edge dots, the optimizer finds it favorable to create a strong potential well in the middle of the chain. This could be interpreted as a way for the system to effectively decouple the edge sites from each other, which is a key requirement for realizing Majorana modes. By creating a strong potential barrier in the middle, the optimizer may be trying to localize the low-energy states at the edges and prevent them from hybridizing with each other through the bulk of the chain. This trend is also visible in the four-dot system, where the optimized on-site energies become highly inhomogeneous, with larger chemical potentials for the middle sites compared to the edge sites. This suggests that even in longer chains, the optimization procedure may favor parameter profiles that enhance localization at the edges by creating potential barriers in the middle of the chain.

4.2 Post-Optimization Diagnostics

After presenting the optimized parameters themselves, the next step is to test whether the corresponding Hamiltonians display the expected signatures of Majorana-like physics. This section should therefore focus on the parity-resolved spectrum and on the local observables introduced in the method chapter. After finding the optimized parameters, we can use them to set up the Hamiltonian for the desired number of quantum dots. After the parameter optimizing scheme, we found promising parameters for the interacting 3-dot system. Using the parameters found in the optimization scheme for $U = 0.1$, we can diagonalize the Hamiltonian and look at the parity-resolved spectrum to create Majorana-like states.

$$H = \sum_{i=1}^{3} \epsilon_i n_i + \sum_{i=1}^{2} t_i (c_i^\dagger c_{i+1} + h.c.) + \sum_{i=1}^{2} \Delta_i (c_i c_{i+1} + h.c.) + U \sum_{i=1}^{3} n_i n_{i+1} \tag{17}$$

Diagonalizing we will obtain $2^3 = 8$ energy levels, which can be grouped into two parity sectors with four levels each. The parameters found are optimized to make the four levels in each parity sector as close to degenerate as possible, which is a key signature of Majorana-like physics. In figure 10 we can see the parity resolved spectrum for the three-dot system at $U = 0.1$. Using these parameters we are able to create a set of Majorana-like states:

$$\gamma_1 = \sum_{i=1}^{n=4} |e\rangle \langle o| + |o\rangle \langle e|$$

$$\gamma_2 = i \sum_{i=1}^{n=4} (|e\rangle \langle o| - |o\rangle \langle e|)$$

with these Majorana operators, we are able to choose the number of energy levels we want to include in the operators, and we are able to explore the properties of these operators and if they are stable under the braiding protocol.

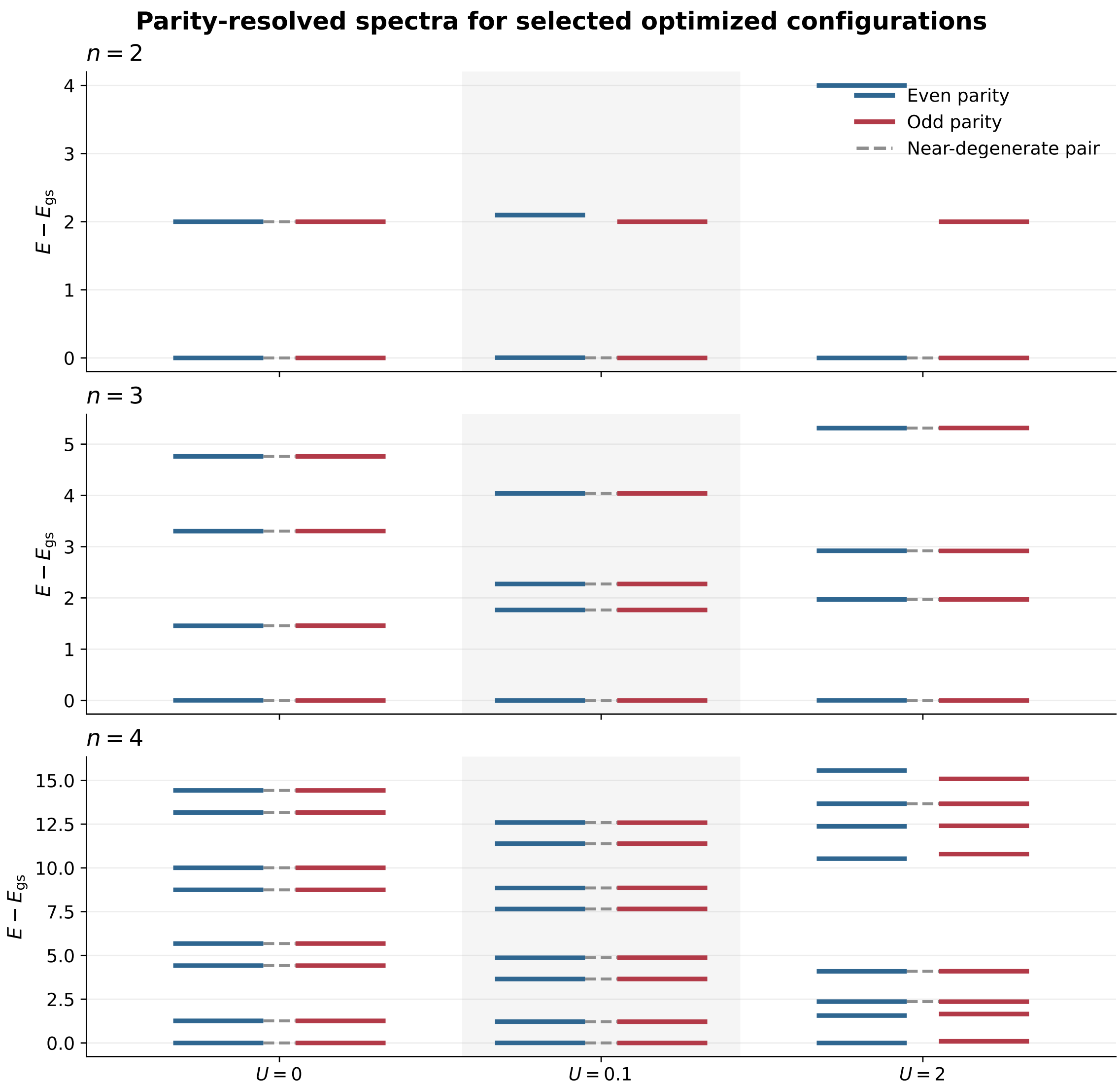

Figure 11: Parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size n, and energies are shifted so that the ground state is at zero.

4.2.1 Parity-resolved spectra and low-energy gaps

Figure 11 shows the parity-resolved spectra for the best optimized configurations at selected interaction strengths. Each panel corresponds to a different system size $n = 2, 3 \text{and} 4$, with interaction strengths set at $U = 0$, $U = 0.1$, and $U = 2$ and energies are shifted so that the ground state is at zero.

We can see that for all system sizes, the optimizer managed to find a set of parameters where the four energy levels in each parity sector are close to degenerate in the non-interacting system. The first sign of instability appears at $U = 0.1$, for the two-dot system, where the degeneracy of the excited state is slightly lifted. This effect becomes more pronounced at stronger interactions, where the excited states are completely split for $U = 2.0$.
The four dot system shows a different trend. The optimizer finds a stable set of parameters for both $U = 0$ and $U = 0.1$, where the four energy levels in each parity sector are close to degenerate and the energy gap is relatively large. However, with stronger interaction strengths, $U = 2.0$, we can see that the optimizer struggled. Even though ground state degeneracy is highly

n	U	δ_0	Δ_{gap}	Q_0	$\overline{\lvert M_{\text{edge}}\rvert}$	$\max\lvert M_{\text{bulk}}\rvert$
2	0	7.64e-05	2	0.00139	1	0
2	0.1	0.00344	2	0.00142	1	0
3	0.1	9.39e-05	1.77	0.0492	0.998	0.000202
4	0.1	0.00423	1.21	0.007	0.986	0.0233

Table 2: Summary of local diagnostics for the representative configurations shown in Fig. 12. Here δ_0 is the splitting of the lowest even-odd pair, Δ_{gap} is the gap separating this pair from the next excited states, Q_0 is the total charge difference between the parity partners, $\overline{|M_{\text{edge}}|}$ is the mean absolute Majorana polarization on the first and last sites, and $\max|M_{\text{bulk}}|$ is the largest absolute polarization on the interior sites.

weighted in the loss function, it did not manage to find a set of parameters where the ground states are close to degenerate. The two lowest states show splitting, and thus these parameters are not ideal for idealizing Majorana modes. This could be due to the fact that the optimization landscape becomes more complex at stronger interactions, and the optimizer may have difficulty finding good solutions in that regime.
Interestingly, the model found a stable set of parameters for all interaction strengths for the three-dot system. Even at $U = 2.0$, the optimizer found a set of parameters where the four energy levels in each parity sector are close to degenerate, although the gap to excited states is smaller than in the non-interacting case. This suggests that for the three-dot system, there may be more flexibility in finding Majorana-like configurations even at stronger interactions, compared to the two-dot and four-dot systems. Due to this, the three-dot system is was chosen as the most promising candidate for the braiding analysis in the next part of the thesis.

4.2.2 Local diagnostics and Majorana character

The parity-resolved spectrum shows whether the optimizer has found a low-energy manifold with approximate even-odd degeneracy, but this is not sufficient on its own to identify Majorana-like states. To test whether the optimized configurations also have the expected spatial structure, I therefore examine local diagnostics of the lowest even-odd pair. The two most useful quantities are the site-resolved Majorana polarization and the local charge difference between the parity partners. A convincing Majorana-like configuration should display opposite polarization on the two ends of the chain, weak polarization in the middle, and only a small charge difference between the even and odd states.

Figure 12 compares the spatial profiles directly, while Table 2 summarizes the corresponding scalar diagnostics. For the two-dot system, the edge polarization is essentially ideal, with $\overline{|M_{\text{edge}}|} \approx 1$ and no bulk contribution by construction. The total charge difference is also very small, $Q_0 \approx 1.4 \times 10^{-3}$, both at $U = 0$ and at $U = 0.1$. These two-dot cases are therefore useful reference points, but because there are no interior sites they do not probe how well the optimizer suppresses Majorana weight in the bulk.

The three-dot configuration at $U = 0.1$ is the clearest interacting example of the expected Majorana pattern. The ground-state splitting is only $\delta_0 = 9.39 \times 10^{-5}$, the excitation gap remains large, $\Delta_{\text{gap}} = 1.77$, and the bulk polarization is almost completely suppressed, with $\max|M_{\text{bulk}}| = 2.02 \times 10^{-4}$. This means that even though the total charge difference is somewhat larger than in the two-dot reference cases, the low-energy pair is still extremely well localized at the ends of the chain. In contrast, the four-dot configuration at $U = 0.1$ has a comparably strong edge polarization, $\overline{|M_{\text{edge}}|} \approx 0.986$, but the splitting increases to $\delta_0 = 4.23 \times 10^{-3}$ and the bulk polarization rises to $\max|M_{\text{bulk}}| = 2.33 \times 10^{-2}$. The mode is therefore less cleanly separated from the interior of the chain, even though the charge difference is smaller than in the three-dot case.

Taken together, the figure and table show that different diagnostics emphasize different

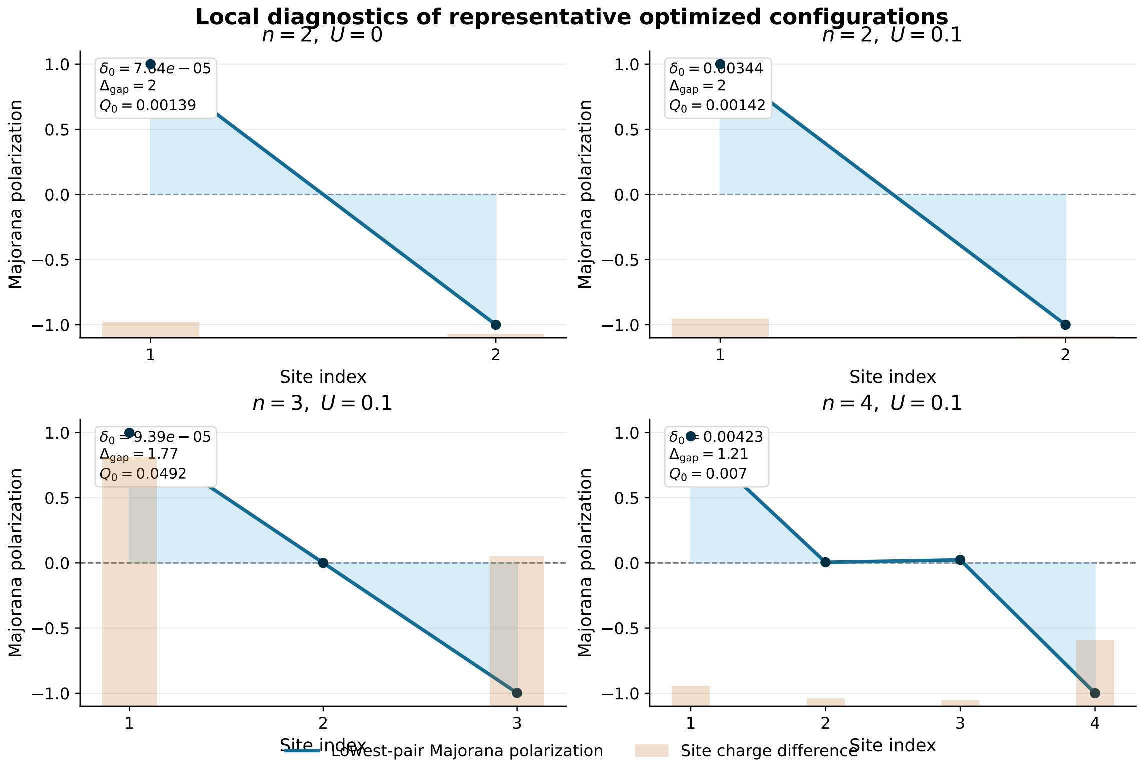

Figure 12: Local diagnostics for representative optimized configurations. The blue curve shows the Majorana polarization profile of the lowest even-odd pair, while the orange bars show the local charge difference $|\langle e|n_i|e\rangle - \langle o|n_i|o\rangle|$ on each site. The inset in each panel reports the ground-state splitting δ_0, the low-energy gap Δ_{gap}, and the total charge difference Q_0. A clean Majorana-like configuration is expected to have opposite edge polarization, minimal bulk weight, and a small charge imbalance between the parity partners.

aspects of the optimized states. The four-dot case performs well in terms of charge neutrality, while the three-dot case performs much better in terms of spectral near-degeneracy and suppression of bulk Majorana weight. Since the subsequent braiding analysis requires a low-energy subspace that is both nearly degenerate and spatially well localized, the interacting three-dot system remains the most convincing candidate for the rest of the thesis.

4.3 Braiding Results in the Effective Model

Before interpreting the results of the full microscopic setup, it is useful to summarize the idealized braiding calculation. This section should therefore act as a reference point for what the intended exchange looks like in the simplified four-Majorana model.

Prior to this section we have found a few different optimized n-dot configurations for the Poor Man's Majorana chain, and found a couple of candidates for the braiding analysis. From 12, 11 and 2 we can see that the three-dot system at $U = 0.1$ is the most promising candidate for the braiding analysis, since it has a very small ground-state splitting, a large gap to excited states, and a very clean Majorana polarization profile.

Using the optimized parameters for the three-dot system at $U = 0.1$, we can set up the Hamiltonian and perform the idealized braiding protocol. The hamiltonian is given by equation 17, and the optimized parameters are $\Delta_1 = 1.001$, $\Delta_2 = 1.002$, $\epsilon_1 = -0.101$, $\epsilon_2 = -0.607$, and $\epsilon_3 = -0.003$. The hopping amplitudes are fixed to $t_i = 1$ from table 1. With these parameters, we diagonalize the Hamiltonian, and sort the 8 different eigenvalues and vectors by energy and by parity. This allows us to construct the Majorana operators γ_1 and γ_2 with any desired number of energy levels included in the sum.

There are three different ways we tested the braiding protocol on our Majorana operators. First was simply to set up the effective Hamiltonian for defined for sweet spot Majorana modes, 13.
This Hamiltonian is defined by 4 Majorana operators, and the couplings between them. Since we wanted to use the optimized parameters for the three-dot system, and we want to replicate the setup of 5, we define it so that in refrence to the figure 5 we get γ_0 and γ_1 from subsystem A, γ_2 and $\tilde{\gamma}_2$ from subsystem B, and γ_3 and $\tilde{\gamma}_3$ from subsystem C. The couplings in between them are created by some simple pulses, which are constructed as a combination of sigmoid functions, which are smooth and have a well defined time derivative. This braiding model assumes a perfect four-Majorana structure, with the parameters at their respective sweet spots, and no additional couplings or perturbations. The purpose of this calculation is to confirm that the intended braiding protocol produces the expected unitary transformation in the idealized limit, and to see if the Majorana operators constructed from the optimized parameters can reproduce this behavior when projected onto the low-energy subspace, $P_g s$, where $P_g s$ is the projection operator onto the ground-state of the full Hamiltonian $H_f ull = H_A + H_B + H_C$.

The next step is to implement the Junction Hamiltonian, 16 & 14, which is defined by the microscopic hopping and pairing terms that couple the different subsystems. This allows us to test whether the optimized microscopic setup can reproduce the intended braiding behavior in its low-energy sector, and whether there is any leakage between different sites during the protocol. This also tells us which one of the γ_2 and $\tilde{\gamma}_2$, as well as which one of the γ_3 and $\tilde{\gamma}_3$ are the ones that are actually participating in the braiding, since the junction Hamiltonian is defined by the microscopic hopping and pairing terms, and not by hand-chosen Majorana bilinears. This method also allows us to explore the weights of the Majoranas during the braiding protocol. What this means is that we can detect if γ_2 and $\tilde{\gamma}_2$, are equally weighted or not. Ideally the weights should heavily favor one of the two, which would indicate that the braiding is happening in a well-defined low-energy subspace. During the protocol, we detected that the Majoranas that

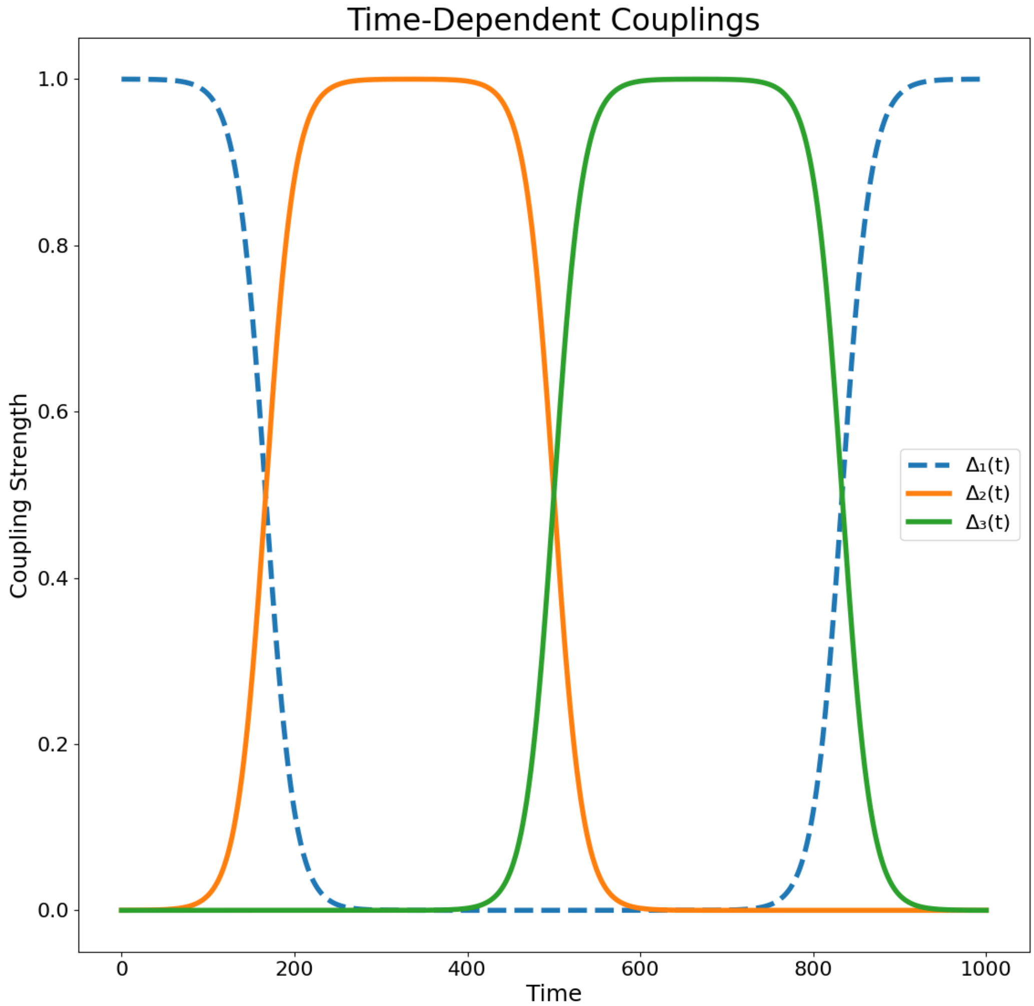

Figure 13: Time-dependent couplings used in the idealized braiding protocol. The left panel shows the couplings $\Delta_{ij}(t)$ between the Majorana modes, while the right panel shows the corresponding instantaneous spectrum of the effective Hamiltonian. The low-energy manifold remains approximately degenerate and well separated from excited states throughout the protocol.

were participating in the braiding were $\tilde{\gamma}_2$ and $\tilde{\gamma}_3$, which are the ones that have the largest weight during the protocol. There was little to no leakage between different sites, which is a good sign that the braiding is happening in a well-defined low-energy subspace.

The last check we performed in order is to substitute γ_2 and γ_3, from 13 with $c^\dagger$ and c for the different subsystems. So the effective Hamiltonian becomes:

$$H_{\text{eff}}(t) = i\Delta_{01}(t)\gamma_0\gamma_1 + i\Delta_{2c}(t)\gamma_0(c_B^\dagger + c_B) + i\Delta_{3c}(t)(c_C^\dagger + c_C) \tag{18}$$

This is also a more realistic model for the braiding protocol, since it does not assume a perfect four-Majorana structure, and it allows us to test whether the optimized parameters can reproduce the intended braiding behavior even when we include the full microscopic details of the junction couplings. This is also a good check to see if the Majorana operators constructed from the optimized parameters can reproduce the expected unitary transformation when projected onto the low-energy subspace, even when we include the full microscopic details of the junction couplings.

4.3.1 Evolution of the effective spectrum

Figure 14 shows the reference braid in the effective four-Majorana model. The three couplings are turned on sequentially, so that the dominant pairing is first between γ_0 and γ_1, then between γ_0 and γ_2, and finally between γ_0 and γ_3 before returning to the initial configuration. Throughout this cycle, the low-energy manifold remains essentially four-fold degenerate. The largest instantaneous splitting within the ground-state manifold is only $\max(E_3 - E_0) = 2.28 \times 10^{-12}$, while the minimum gap to the excited states remains open at $\min(E_4 - E_3) = 1.414$.

This is the behavior expected from an ideal adiabatic braid. The low-energy states stay well isolated from the rest of the spectrum, so the Kato transport can be applied without the path ever closing the many-body gap. In this sense, Fig. 14 provides a clean numerical benchmark for what the microscopic braid should reproduce as closely as possible.

4.3.2 Berry phase and ideal exchange behavior

Using the Kato formulation of adiabatic transport, I compute the unitary acting on the four-dimensional low-energy subspace after one braid cycle. In the effective model, the transported operator remains highly unitary, with $\|U^\dagger U - I\| = 3.96 \times 10^{-14}$, and the transported projector agrees with the final ground-state subspace up to a transport error of 6.67×10^{-6}. The resulting braid also reproduces the expected single-exchange action on the Majorana operators, with the largest error in the transformations $\gamma_2 \to -\gamma_3$ and $\gamma_3 \to \gamma_2$ equal to 1.33×10^{-5}.

The same conclusion is visible in the parity-resolved gate blocks. After diagonalizing total parity within the ground-state manifold, the odd and even 2×2 blocks agree with the target exchange gate to within 6.53×10^{-8}, while the off-block leakage is only 1.02×10^{-11}. The effective model therefore behaves as a nearly ideal non-Abelian exchange, and these numbers serve as a reference point for the more realistic projected calculations below.

4.4 Braiding Results for Optimized Majorana Modes

This section should present the main braiding results of the thesis: whether the optimized microscopic system can reproduce the exchange behavior suggested by the ideal model.

4.4.1 Projected microscopic braid

After replacing the hand-chosen couplings of the toy model by projected microscopic junction operators, the braid remains remarkably close to the ideal reference. Figure 15 shows that the

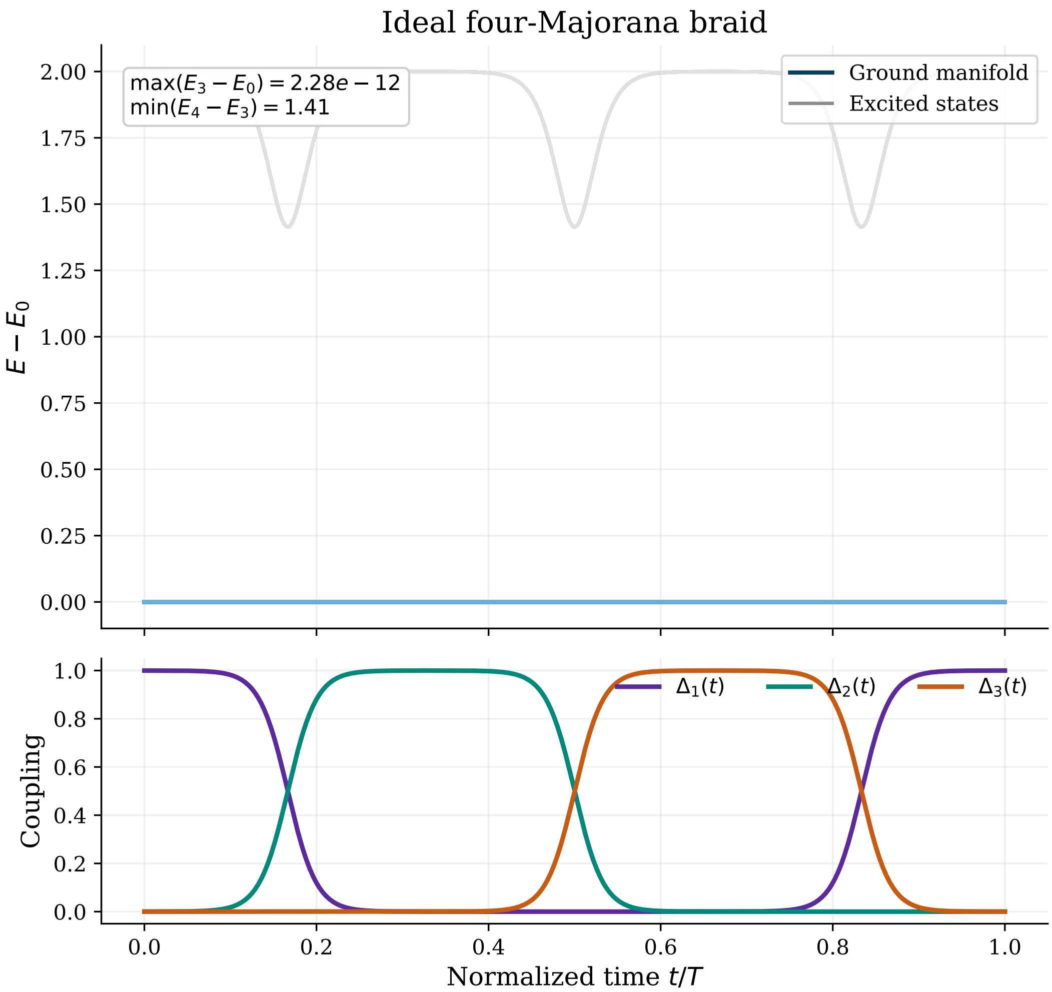

Figure 14: Idealized four-Majorana braiding protocol. The top panel shows the instantaneous spectrum of the effective Hamiltonian during the braid, with all energies shifted so that the lowest state remains at zero. The bottom panel shows the three smooth pulse couplings that move the dominant Majorana pairing through the network.

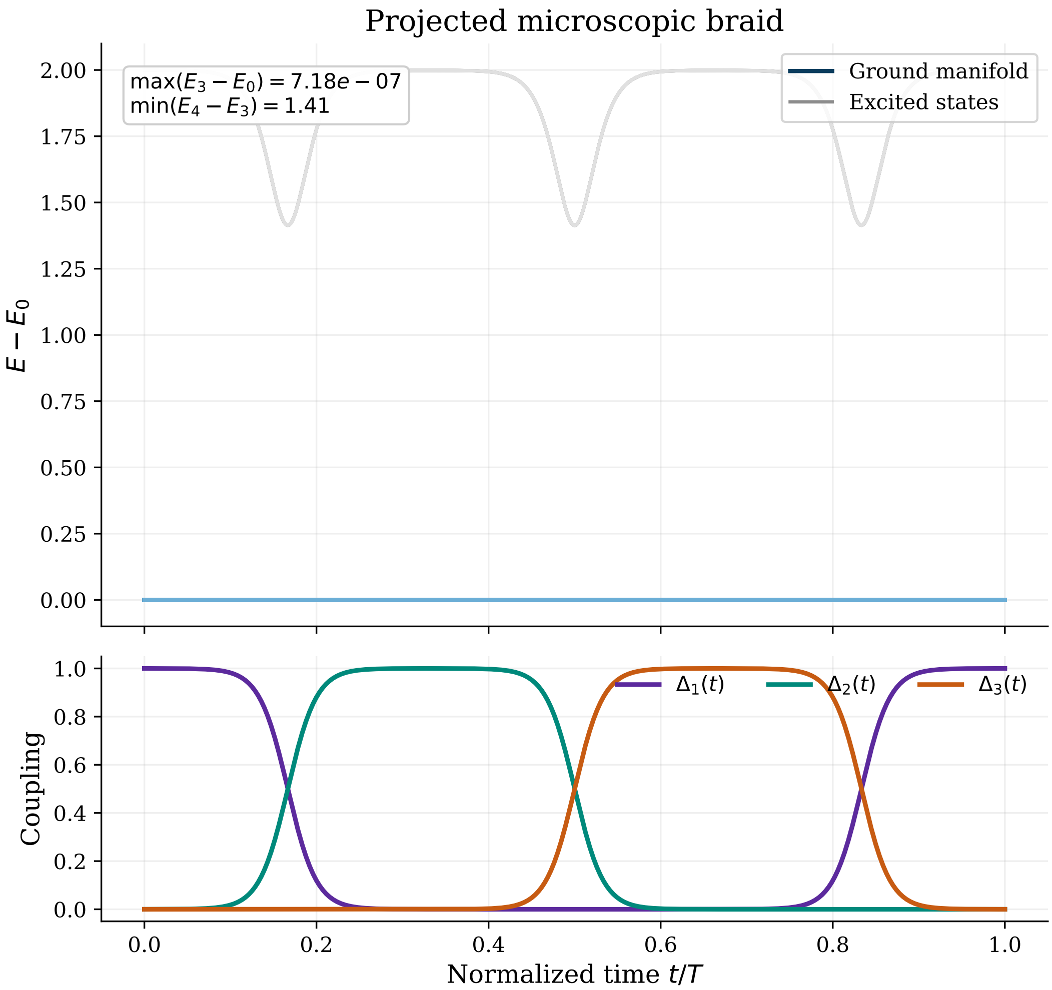

Figure 15: Braiding protocol after replacing the idealized couplings by projected microscopic junction operators. The spectral evolution remains almost identical to the effective-model benchmark, with a well-isolated low-energy manifold throughout the cycle.

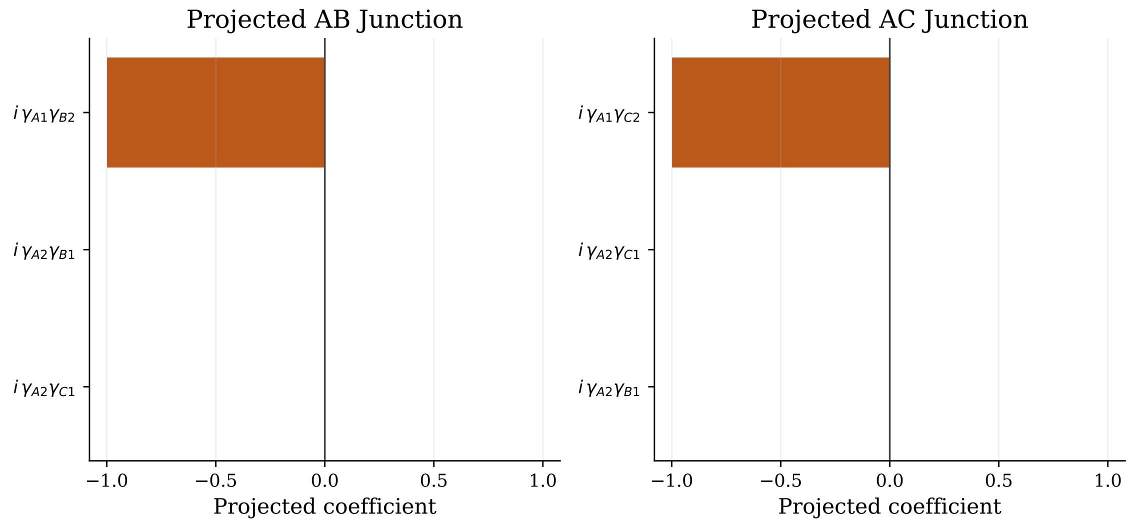

Figure 16: Dominant Majorana-bilinear components of the projected microscopic junction operators. The AB junction is almost entirely $i\gamma_{A1}\gamma_{B2}$, while the AC junction is almost entirely $i\gamma_{A1}\gamma_{C2}$. All subleading bilinears are strongly suppressed.

four-dimensional low-energy manifold is still well separated from higher states throughout the cycle. The maximum splitting inside the ground-state manifold increases only to $\max(E_3-E_0) = 7.18 \times 10^{-7}$, while the minimum gap remains large, $\min(E_4 - E_3) = 1.413$. The projected microscopic protocol therefore preserves the same basic adiabatic structure as the idealized model.

The junction decomposition in Fig. 16 makes the physical meaning of this agreement more transparent. The projected AB coupling is dominated by the bilinear $i\gamma_{A1}\gamma_{B2}$ with coefficient -0.999, and the projected AC coupling is dominated by $i\gamma_{A1}\gamma_{C2}$ with the same coefficient. This identifies γ_{B2} and γ_{C2} as the Majoranas that actually participate in the microscopic braid, while γ_{A1} acts as the shared Majorana through which the exchange is mediated. The microscopic model is therefore not just qualitatively similar to the toy model: after projection, the junction operators line up almost perfectly with the intended effective bilinears.

4.4.2 Gate-validation checks

Model	$\max(E_3-E_0)$	$\min(E_4-E_3)$	Max 1x err.	Leakage	Odd err.	Even err.
Ideal effective model	2.275e-12	1.414	1.334e-05	1.016e-11	6.535e-08	6.535e-08
Projected microscopic junction model	7.175e-07	1.413	1.334e-05	8.844e-16	6.538e-08	6.538e-08
Projected local-operator model	4.663e-15	1.414	1.334e-05	7.287e-16	6.537e-08	6.537e-08

Table 3: Comparison of the main braid diagnostics for the ideal effective model, the projected microscopic junction model, and the alternative projected local-operator model. Here $\max(E_3-E_0)$ measures the largest instantaneous splitting inside the four-dimensional low-energy manifold, $\min(E_4-E_3)$ is the minimum gap to the excited sector, "Max 1x err." is the largest deviation in the expected single-exchange Majorana map, "Leakage" is the off-block norm in the parity basis, and the last two columns compare the odd and even parity blocks with the target exchange gate.

Table 3 shows that the microscopic braid reproduces the ideal benchmark extremely well. The largest single-exchange error stays at 1.33×10^{-5} in all three models, and the odd and even parity blocks continue to match the target gate to approximately 6.54×10^{-8}. In other words, once the microscopic junction operators are projected into the low-energy manifold, they generate essentially the same braid unitary as the ideal four-Majorana model. The projected

Projected operator	Dominant Majorana	Coefficient
$\chi_{B1,y}$	γ_{B2}	-1.000
$\chi_{B2,y}$	γ_{B2}	1.000
$\chi_{C1,y}$	γ_{C2}	-1.000
$\chi_{C2,y}$	γ_{C2}	1.000
Pairwise relation errors		
$\chi_{B1,y} \approx -\chi_{B2,y}$	opposite sign	8.969e-14
$\chi_{C1,y} \approx -\chi_{C2,y}$	opposite sign	5.787e-14

Table 4: Projected local-site operators used in the alternative braid model. The table shows that the y-type local operators on the first and last sites of subsystems B and C collapse onto the same low-energy Majorana channels, γ_{B2} and γ_{C2}, up to an overall sign.

local-operator version produces the same conclusion, which makes it a useful consistency check rather than an independent competing protocol.

Table 4 also explains why the local-site version in `braiding_model_ad2.py` appears insensitive to whether the first or last site of subsystem B or C is used. After projection, both $\chi_{B1,y}$ and $\chi_{B2,y}$ reduce to γ_{B2} up to opposite sign, and similarly both $\chi_{C1,y}$ and $\chi_{C2,y}$ reduce to γ_{C2}. The different microscopic site choices therefore feed into the same low-energy braid channel rather than producing genuinely different braiding operations. This makes the apparent insensitivity of the projected local model a physical consequence of the low-energy projection, not a numerical inconsistency in the implementation.

4.5 Braiding in Higher Energy Levels

The previous braiding checks focused on the lowest projected manifold. To test how much of that agreement survives once excited sectors are retained, I repeated the braid diagnostics for projection dimensions $\dim P = 8, 32, 56, 80, 256$, and 512, and for interaction strengths $U = 0$, 0.1, and 2.0. For each projected Hamiltonian I choose the transported band from the largest persistent spectral gap along the braid path. This separates the projection dimension, which controls how much Hilbert space is retained, from the transported dimension $\dim T$, which controls which isolated band is actually braided.

The physical junction is taken to be the projected local minus operator, $i(c^\dagger - c)$, on the inner B and C sites. A subtle point is that there are now two possible braid targets. The state-constructed target uses the many-body operators γ_2 and γ_3 obtained from the paired even and odd eigenstates, while the microscopic target uses the projected analytical Jordan–Wigner operators Γ_2^P and Γ_3^P. Comparing the physical braid only with the state-constructed target can therefore mix two questions: whether the braid succeeds, and whether the state-constructed Majoranas coincide with the projected analytical Majoranas.

Table 5 shows that the apparent disagreement is dominated by the target mismatch. For $U = 0$, the state target and projected analytical target differ only weakly through the moderate projections, and the physical braid matches the analytical target at the 10^{-6}–10^{-4} level. For $U = 0.1$, the target mismatch grows from a few percent to order one at the largest projection. For $U = 2$, the mismatch is already order one once excited sectors are included, which means that the state-constructed Majoranas and the projected microscopic Majoranas no longer describe the same braid channel.

Figures 17–19 replace the earlier physical-versus-ideal exchange diagnostics as the main higher-energy braid test. The earlier ideal braid remained nearly perfect because its Hamiltonian and target were built from the same projected Majorana operators. The present diagnostic is less self-referential: it asks whether the microscopic projected operators Γ_2^P and Γ_3^P still represent the same exchange as the state-constructed many-body Majoranas. This gives the expected physical trend, with good agreement at $U = 0$, partial degradation at $U = 0.1$, and a clear

U	$\dim P$	$\dim T$	Phys. vs. γ target	Phys. vs. Γ^P target	Target mismatch	Γ^P algebra
0	8	4	2.14e-03	1.15e-06	2.14e-03	1.65e-15
0	32	8	4.27e-03	9.80e-05	4.27e-03	1.08e-04
0	56	8	1.49e-02	1.36e-04	1.49e-02	7.57e-05
0	80	12	4.27e-03	1.10e-04	4.27e-03	1.06e-04
0	256	16	4.98e-03	1.32e-04	4.98e-03	8.44e-05
0	512	16	2.90e-01	1.52e-04	2.90e-01	2.99e-15
0.1	8	4	3.07e-02	9.46e-07	3.07e-02	3.20e-15
0.1	32	8	5.17e-02	1.59e-04	5.17e-02	3.67e-04
0.1	56	12	3.32e-02	2.67e-04	3.32e-02	5.90e-04
0.1	80	12	5.49e-02	2.84e-04	5.49e-02	5.82e-04
0.1	256	16	9.67e-02	4.24e-04	9.67e-02	5.99e-04
0.1	512	16	1.00e+00	5.92e-04	9.99e-01	3.22e-15
2	8	4	4.28e-01	1.02e-06	4.28e-01	1.95e-15
2	32	8	5.48e-01	2.10e-02	5.53e-01	5.77e-02
2	56	12	5.51e-01	5.10e-02	5.66e-01	1.75e-01
2	80	12	7.15e-01	5.37e-02	7.33e-01	1.53e-01
2	256	16	9.27e-01	4.09e-02	9.40e-01	1.50e-01
2	512	16	7.25e-01	2.04e-02	7.15e-01	3.15e-15

Table 5: Target diagnostic for the corrected higher-energy braid scan with $\dim P = 8, 32, 56, 80, 256, 512$. All errors are normalized in the transported band. “Phys. vs. γ target” compares the physical braid to $\exp(-\pi\gamma_2\gamma_3/4)$, where γ_2 and γ_3 are constructed from the paired eigenstates. “Phys. vs. Γ^P target” compares the same physical braid to $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$, where Γ_2^P and Γ_3^P are projected analytical Jordan–Wigner Majoranas. “Target mismatch” directly compares these two target gates.

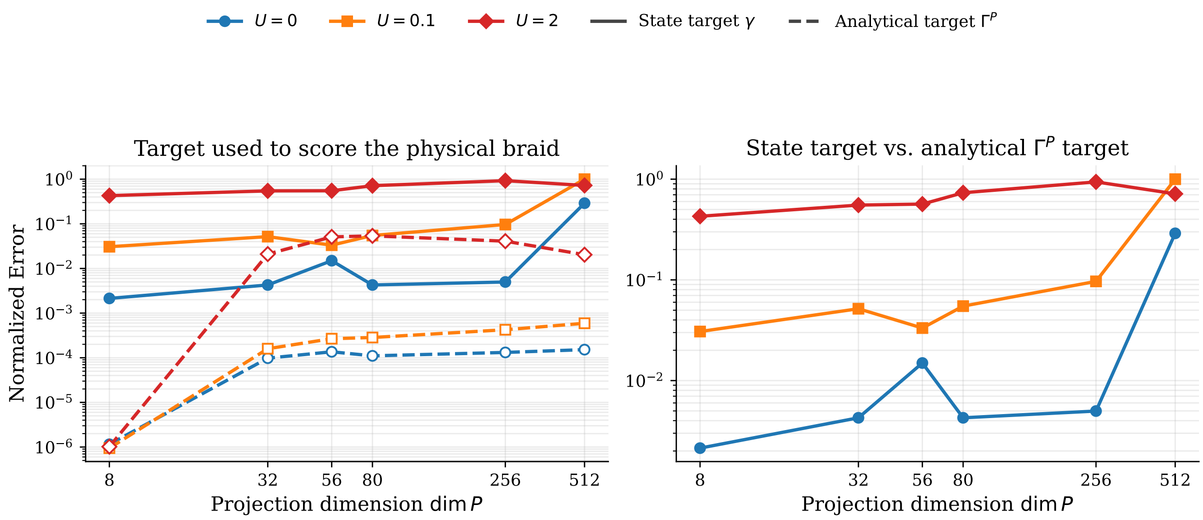

Figure 17: Comparison of the two target gates used to score the physical braid. The solid curves in the left panel compare the physical braid to the state-constructed target $\exp(-\pi\gamma_2\gamma_3/4)$, while the dashed curves compare it to the projected analytical target $\exp(-\pi\Gamma_2^P\Gamma_3^P/4)$. The right panel shows the direct mismatch between these two target gates. The hierarchy is clear: $U = 0$ remains close, $U = 0.1$ is intermediate, and $U = 2$ separates strongly from the analytical Majorana description.

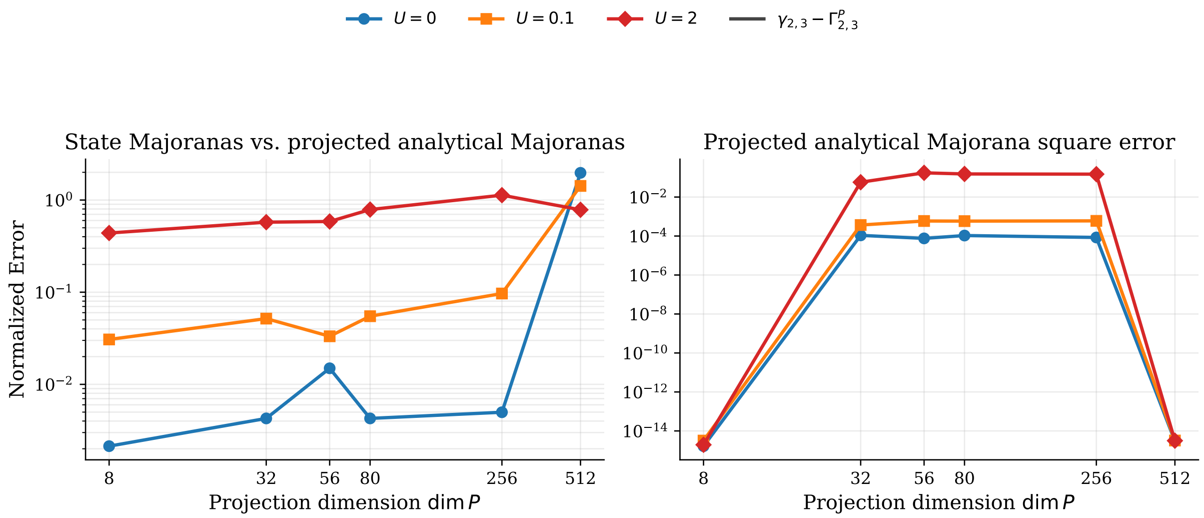

Figure 18: Direct operator diagnostic behind the target mismatch. The left panel compares the state-constructed Majoranas with the projected analytical Majoranas inside the transported band. The B and C mismatches are equal by symmetry, so a single $\gamma_{2,3} - \Gamma^P_{2,3}$ curve is shown. The right panel shows the maximum square error of the projected analytical Majoranas. The strong-interaction case has both a large operator mismatch and a visibly degraded projected Majorana algebra once excited sectors are retained.

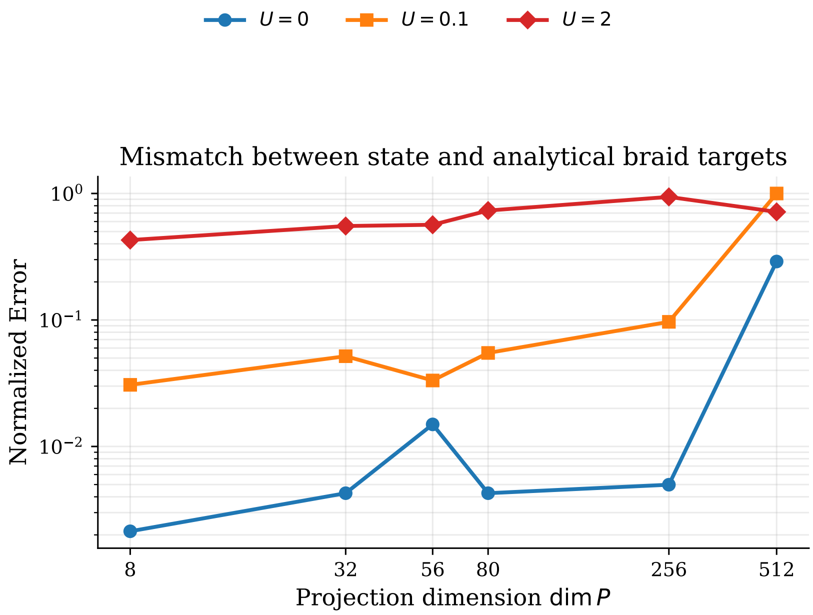

Figure 19: Interaction hierarchy extracted from the direct target mismatch. This is the most compact version of the diagnostic: noninteracting braids remain close to the analytical Majorana target, weak interactions produce a modest drift, and strong interactions produce an order-one separation between the state-constructed and microscopic braid targets.

breakdown at $U = 2$.

5 Discussion

This section interprets the numerical results in relation to the main goal of the thesis: to test whether optimized interacting quantum-dot chains can support Majorana-like low-energy physics and whether that physics remains useful for braiding. The results suggest that the optimization procedure does more than recover known non-interacting limits. It also identifies nontrivial inhomogeneous parameter profiles that still preserve the low-energy signatures needed for a Majorana interpretation.

5.1 What the optimized parameter profiles imply

One of the clearest trends in the optimization results is that the best interacting configurations do not remain close to a simple homogeneous sweet-spot picture. For the two-dot system, the optimizer reproduces the expected non-interacting limit at $U = 0$, which is an important sanity check. As the interaction strength increases, however, the loss grows and the optimized parameters are pushed away from that point. This should not be interpreted as meaning that interacting two-dot systems cannot retain any useful degeneracy. Rather, the more precise conclusion is that the loss function used here targets approximate degeneracy across all parity-resolved sectors, while the familiar interacting two-dot picture only guarantees the ground-state degeneracy.

For the three- and four-dot systems, the optimizer repeatedly favors strong inhomogeneity in the middle of the chain. In particular, the middle-site chemical potentials become much larger in magnitude than those on the edges, and in the four-dot system the central pairing amplitudes are often enhanced as well. A natural interpretation is that the optimizer is trying to reduce hybridization between the edge modes by engineering an effective barrier or filtering region in the center of the chain. This interpretation is consistent with the Majorana-polarization profiles and with the parity-resolved spectra, both of which indicate that low-energy weight remains concentrated near the outer sites in the best weakly interacting configurations.

At the same time, this trend should be treated with some caution. Because the hopping amplitudes were fixed to $t_i = 1$, the optimizer could only respond to stronger interactions by reshaping Δ_i and ϵ_i. The repeated appearance of large middle-site potentials may therefore be a genuine physical feature, but it may also partly reflect the restrictions built into the ansatz. A useful next step would be to test whether the same tendency appears when the hopping amplitudes are allowed to vary under a constraint that prevents trivial global rescaling.

5.2 What the braiding results imply

The braiding analysis shows a clear separation between low-energy success and higher-energy breakdown. In the lowest projected manifold, the effective Majorana braid and the projected microscopic junction braid agree extremely well. This is a strong result, because it shows that the optimized microscopic model is not merely producing a vaguely similar spectrum. After projection, the relevant junction operators line up with the intended effective Majorana bilinears closely enough to reproduce the expected exchange gate with very small error.

The higher-energy projection scan sharpens this conclusion. For $U = 0$ and $U = 0.1$, the physical projected-junction braid remains close to the ideal control even as additional excited blocks are included. For $U = 2$, by contrast, the target-gate and exchange-channel errors increase rapidly once the projection space extends beyond the ground-state manifold. The important point is that this deterioration is not uniform across every projected operator. The dominant breakdown occurs in the exchanged channels, while the spectator channels remain much more stable. Physically, this suggests that strong interactions first destroy the clean exchange structure rather than the entire low-energy algebra at once.

The basis-diagnostic plots are also important for interpreting the braiding data correctly. Some of the very large error values that appear when the physical evolution is scored against the ideal projected-Majorana basis do not correspond to equally large physical gate failure. They mainly indicate that, in enlarged projection spaces, the projected microscopic operators are no longer identical to the ideal projected Majoranas used in the control calculation. This distinction matters for the interpretation of the results: the low-energy braid can remain physically meaningful even when comparisons to the ideal basis become misleadingly large.

Taken together, the braiding results support a practical design principle for the optimized PMM chains. Useful braiding behavior appears to require that the tunneling and pairing scales remain large compared with the interaction scale, so that the exchange is protected within a well-isolated low-energy manifold. In that sense, the results do not show that interactions are incompatible with braiding. Instead, they show that interactions strongly constrain the regime in which the microscopic model still realizes the intended Majorana exchange.

5.3 Limitations and outlook

The main limitation of the present study is that the optimization procedure does not locate exact sweet spots; it finds numerically favorable minima of the chosen loss function. In a complicated loss landscape, a value such as 10^{-3} may look essentially optimal even when a better solution exists elsewhere. Because all later diagnostics build on the optimized parameter sets, the braiding results should therefore be interpreted as evidence for approximate and practically useful Majorana-like behavior, not as proof of exact topological protection.

A second limitation is that the present braiding analysis is still highly controlled. The clean agreement found in the lowest manifold relies on projection to a small low-energy space and on an idealized, disorder-free setting. The study does not yet include disorder, quasiparticle poisoning, nonadiabatic driving errors, or explicit coupling to thermal environments. Those effects are precisely the kinds of ingredients that would determine whether the optimized configurations remain useful in a more realistic device-level setting.

Several natural extensions follow from this. On the optimization side, it would be valuable to test more global search strategies, alternative loss functions, and less restrictive parameter ansätze. On the braiding side, it would be interesting to study how disorder and noise affect the projected junction operators, and whether the weakly interacting configurations remain robust when the protocol is pushed beyond the ideal adiabatic limit. The higher-energy results also point to a more specific open question: to what extent can one deliberately optimize not only the ground-state manifold, but also nearby excited manifolds, in order to make the exchange protocol more tolerant to thermal occupation?

6 Conclusion

This thesis investigated whether interacting quantum-dot chains can be optimized to realize Majorana-like low-energy states and whether those optimized configurations support a useful braiding protocol. The optimization reproduces the known two-dot non-interacting limit and, for longer chains, finds more inhomogeneous parameter profiles with strong central structure in the on-site energies and pairing amplitudes. These configurations still display the main low-energy signatures targeted in the study, including near even-odd degeneracy, parity-resolved low-energy structure, and edge-weighted Majorana polarization.

The braiding analysis shows that the agreement between the ideal effective model and the microscopic projected model is excellent inside the lowest projected manifold. For weak interactions, especially at $U = 0.1$, this agreement remains good even when additional low-energy blocks are included. For stronger interactions, the braid deteriorates first in the exchanged channels and then more broadly as higher-energy sectors are added. The results therefore point to a clear physical conclusion: useful braiding is achievable in the optimized model, but only when the relevant pairing and tunneling scales remain sufficiently large compared with the interaction strength and the dynamics stay confined to a well-isolated low-energy subspace.

Overall, the thesis supports the view that optimized PMM chains can provide a meaningful microscopic route toward Majorana-based braiding, while also showing that the quality of the braid is highly sensitive to how interactions reshape the low-energy manifold. The most natural next steps are to improve the optimization strategy, test less restricted parameter families, and include disorder, noise, and finite-temperature effects in the braiding analysis.